

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

STRUCTURAL ISOMERISM

General Classification of Isomerism

1. Two isomeric forms of a saturated hydrocarbon:

(A) Have the same structure
 (B) Have different compositions of elements
 (C) Have the same molecular formula
 (D) All the these are correct

CSI069

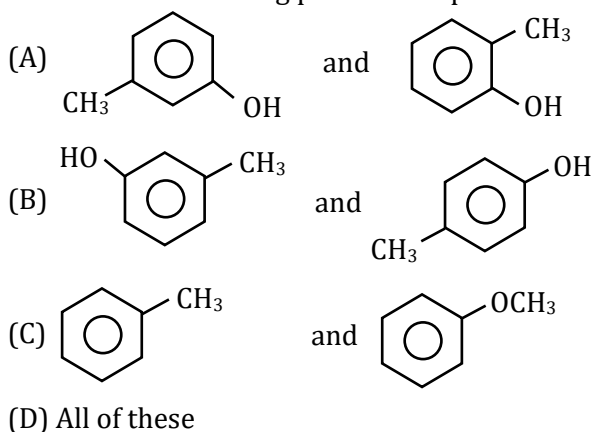
Position Isomerism

2. Positional isomers of 1-bromo-1-chlorocyclohexane are:

(A) 3 (B) 5 (C) 4 (D) 6

CSI062

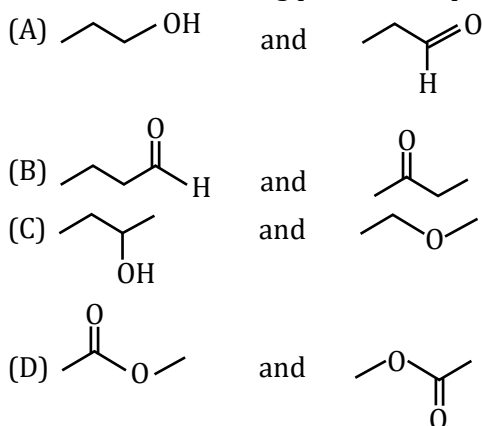
3. Which of the following pairs of compounds are not positional isomers?



CSI063

Functional Isomerism

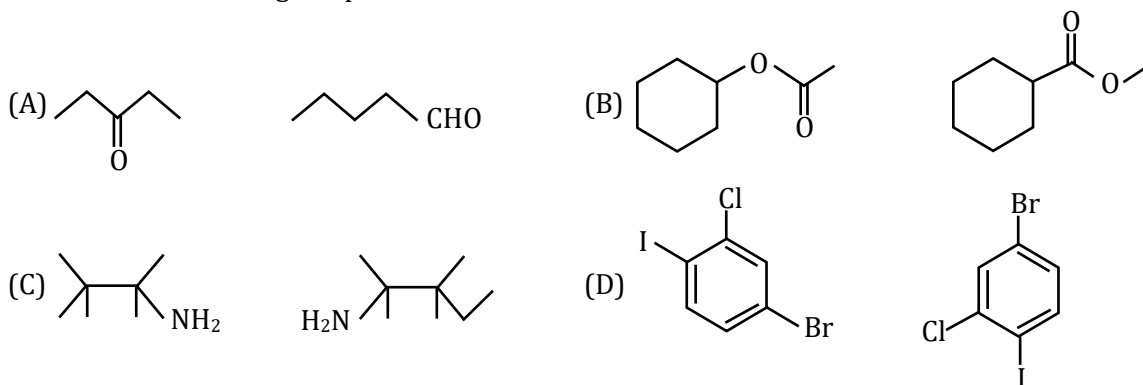
4. Which of the following pairs of compounds are functional isomers?



CSI065

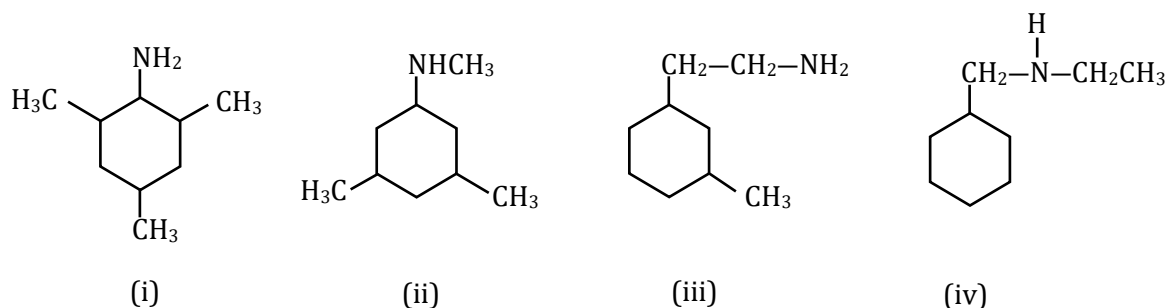
Metamerism

5. Which of the following is a pair of metamers?



CS1066

6. Which of the following is **not the CORRECT** relationship?



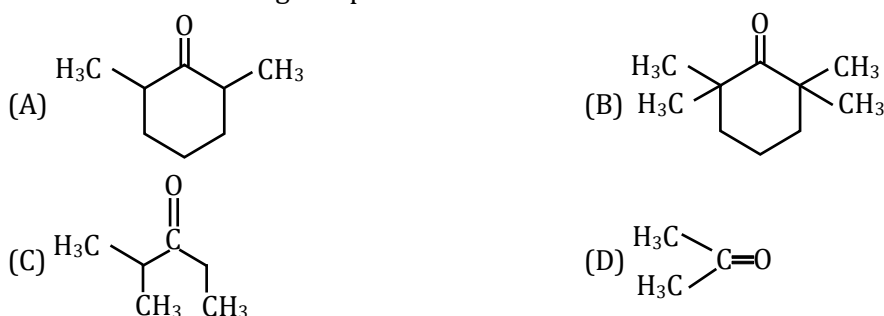
(A) (i) and (ii) are functional isomers
(C) (i) and (iv) are positional isomers

(B) (ii) and (iv) are metamers
(D) (i) and (iii) are chain isomers

CS1071

Tautomerism and its Mechanism

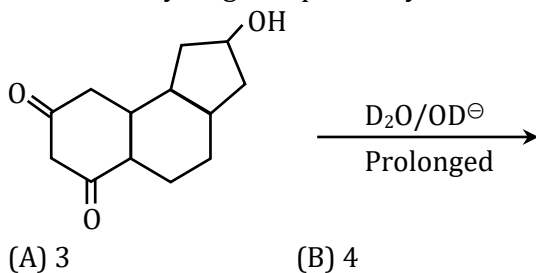
7. Which of the following compounds cannot exhibit keto-enol tautomerism?



CS1075

Enol Content and Kinetic Isotopic Effect

8. Number of hydrogen replaced by Deuterium in the given reaction are :



CS1076

Identification and Number of Structural Isomers

9. How many different dibromophenols are possible?
 (A) 8 (B) 7 (C) 6 (D) 5
10. Number of structural isomers of C_3H_9N are :
 (A) 6 (B) 5 (C) 4 (D) 3

CS1067

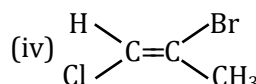
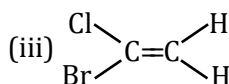
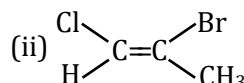
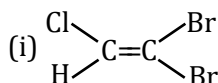
CS1072

GEOMETRICAL ISOMERISM AND CONFORMATIONS**General Definition of Stereoisomerism, Conditions of Geometrical Isomerism in Alkene**

11. Which of the following compound give geometrical isomerism?
 (A) CH_2Cl_2 (B) CH_2Cl-CH_2Cl (C) $CHBr = CHCl$ (D) CH_2Cl-CH_2Br

CGI072

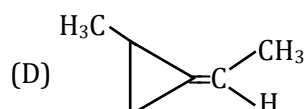
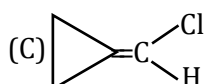
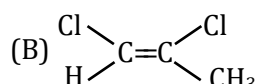
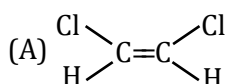
12. Which is a pair of geometrical isomers?



- (A) I and II (B) I and III (C) II and IV (D) III and IV

CGI073

13. Which Compound don't show Geometrical isomerism:



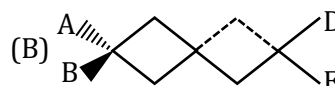
CGI078

14. When isomers have the same structural formula but differ in relative arrangement of atoms or groups are called
 (A) Mesomers (B) Stereoisomers (C) Both A and B (D) None of these

CGI115

Geometrical Isomerism in Allenes, Spiro Compounds, Biphenyls and Triphenyls

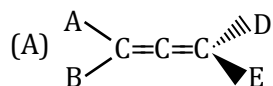
15. Which compound does not show G.I. -



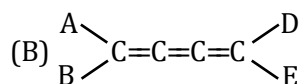
(D) None of these

CGI079

16. Which compound show G.I. -



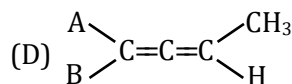
(C) Both (A) & (B)



(D) None of these

CGI080

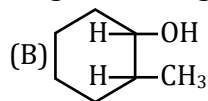
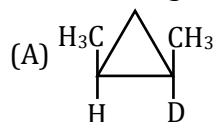
17. Which compound show G.I. -



CGI081

Geometrical Isomerism in Cycloalkanes and Cycloalkenes

18. Which among the following will show geometrical isomerism :

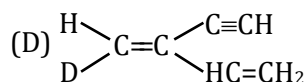
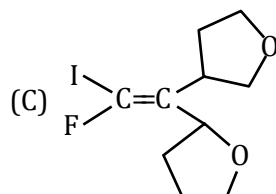
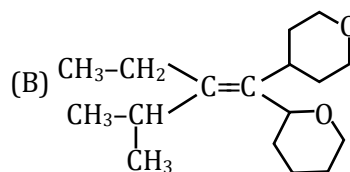
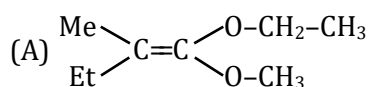


(C) $C_6H_5-CH=N-OH$ (D) All of these

CGI082

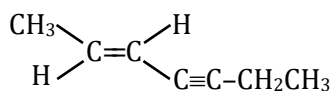
Nomenclature of Geometrical Isomers, Cis-Trans Nomenclature, CIP Rules, E-Z Nomenclature and Syn-Anti Nomenclature

19. Which of the following represent Z-isomer?



CGI088

20. The IUPAC name of the following compounds is:



(A) (Z) - 5 hepten - 3 - yne

(B) (Z) - 2 hepten - 4 - yne

(C) (E) - 5 hepten - 3 - yne

(D) (E) - 2 hepten - 4 - yne

CGI090

Physical Properties of Geometrical Isomerism

21. Which structure have zero dipole moment

(A) cis-2-butene

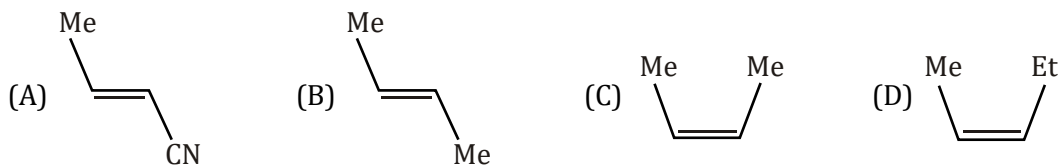
(B) 1, 2-dichloro cyclohexane

(C) p-dichloro benzene

(D) None of these

CGI096

22. Which of the following will have lowest dipole moment?



CGI097

Pseudo Geometrical Isomerism Unit and Calculation of Total Geometrical Isomers

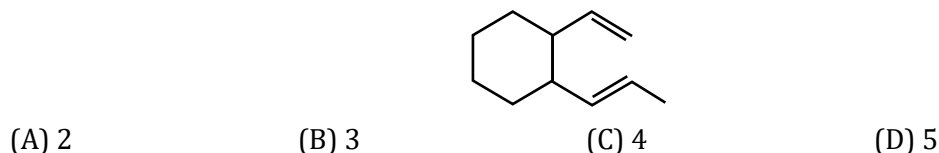
23. The number of geometrical isomers in case of a compound with the structure



- (A) 4 (B) 3 (C) 2 (D) 5

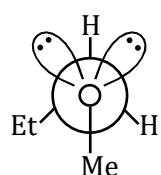
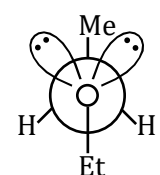
CGI094

24. The number of Geometrical isomer in case of a compound with the structure



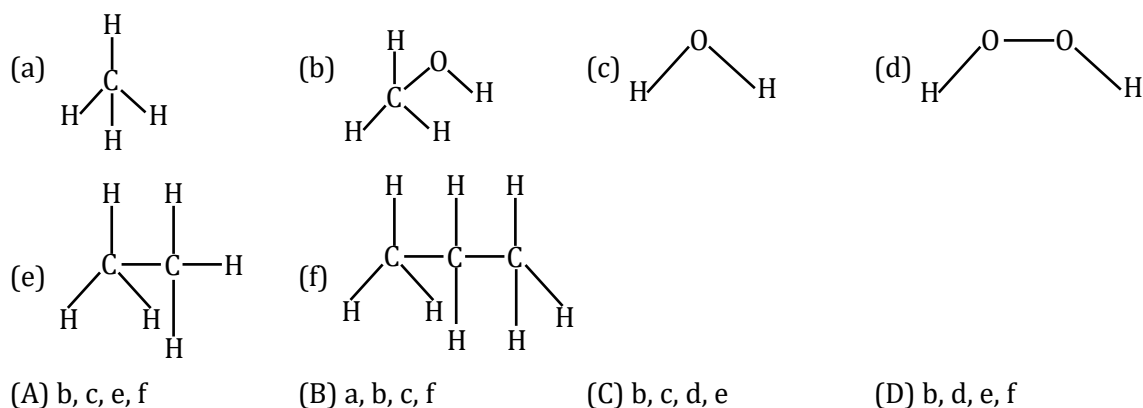
CGI095

Definition of Conformations and Definitions of Gauche, Staggered and Eclipsed

25.  and  are
- (A) Chain isomers (B) Metamers
(C) Positional isomers (D) Conformers

CGI109

26. Which of the following shows conformational isomerism.



CGI111

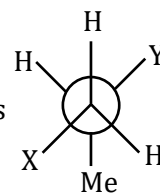
Questions on Stability and Dipole moment

27. Maximum potential energy is associated with which of the following conformers of n-butane:

- (A) Anti (B) Gauche (C) Eclipsed (D) Partial eclipsed

CGI105

28. Newmann projection representation of $\text{CH}_3\text{-CH}_2\text{-CH-Et}$ about $\text{C}_2\text{-C}_3$ is



Then X & Y are respectively:

- (A) Me, Et (B) H, Et (C) Et, H (D) Et, Me

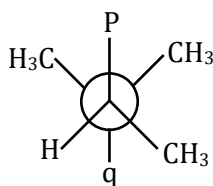
CGI106

29. Bond angle required to acquire most stable form from least stable form in n-butane is :

- (A) 360° (B) 180° (C) 120° (D) 240°

CGI114

30. What is the value of p & q in the following conformer of 2,3-dimethyl butane?



- (A) p = H, q = H (B) p = CH₃, q = CH₃ (C) p = CH₃, q = H (D) p = H, q = CH₃

CGI116

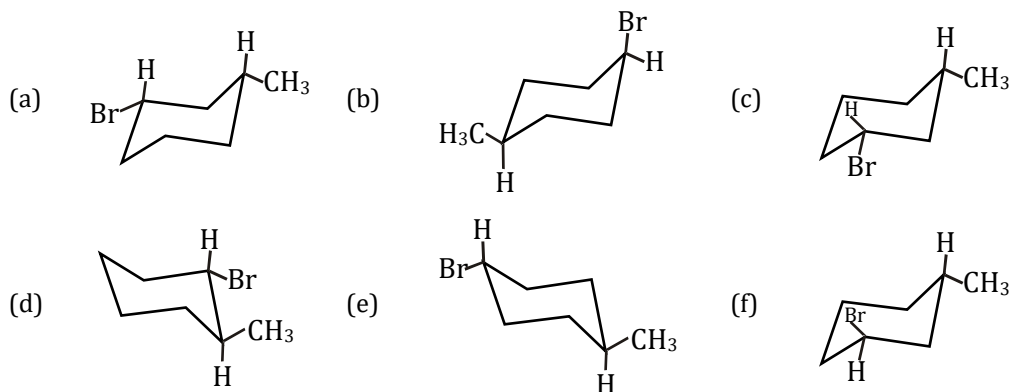
31. Number of possible conformations of n-butane is:

- (A) 2 (B) 4 (C) 6 (D) Infinite

CGI118

Axial-Equatorial orientation in Cyclo hexane, Ring Flipping; Keq of Flipping; Geometrical Isomerism in Cyclohexane and 1;3 Diaxial Repulsion

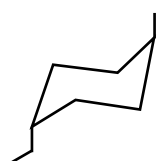
32. How many of the following compounds are cis isomer or a trans isomer.



- (A) 4-cis and 2-trans (B) 3-cis and 3-trans (C) 5-cis and 1-trans (D) 2-cis and 4-trans

CGI120

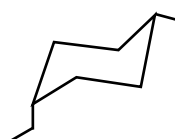
33. Which of the following represent correct stability order of



(I)



(II)



(III)

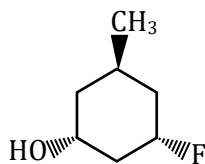


(IV)

- (A) II > I > IV > III (B) II > III > IV > I (C) II > IV > III > I (D) II > IV > I > III

CGI123

34.



In the most stable conformation of the above compound attached groups OH, F & CH₃ are at respectively :

(A) a a a

(B) a a e

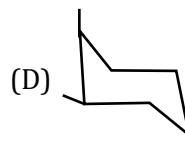
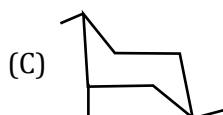
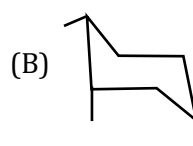
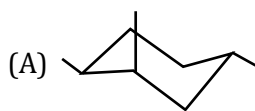
(C) e e e

(D) e e a

CGI124

35.

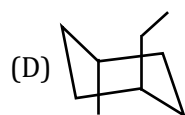
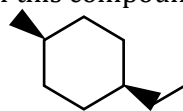
Which of the structures (a-d) will be produced if a "ring flip" occurs in the following compound in chair form?



CGI126

36.

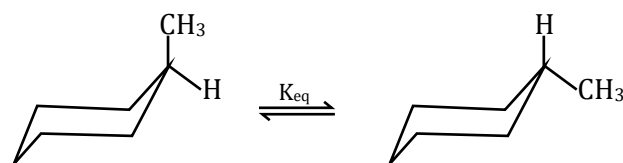
Which is the most stable chair form of this compound?



CGI128

37.

For given reaction

(A) $K_{eq} > 1$ (C) $K_{eq} = 1$ (B) $K_{eq} < 1$

(D) None of these

CGI130

38.

If cis 1, 2-dichlorocyclohexane goes to ring flipping then after ring flipping it will becomes :

(A) cis 1,2-dichlorocyclohexane

(B) Trans 1, 2-dichlorocyclohexane

(C) Both cis, trans

(D) None of these

CGI131

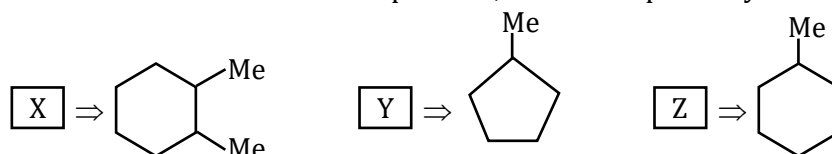
OPTICAL ISOMERISM

Chiral Compound and Chiral Centre and asymmetric centre and pseudo chiral centre

39. Which of the following compounds is not chiral?
 (A) $\text{DCH}_2\text{CH}_2\text{CH}_2\text{Cl}$ (B) $\text{CH}_3\text{CHDCH}_2\text{Cl}$ (C) $\text{CH}_3\text{CHClCH}_2\text{D}$ (D) $\text{CH}_3\text{CH}_2\text{CHDCl}$

COI101

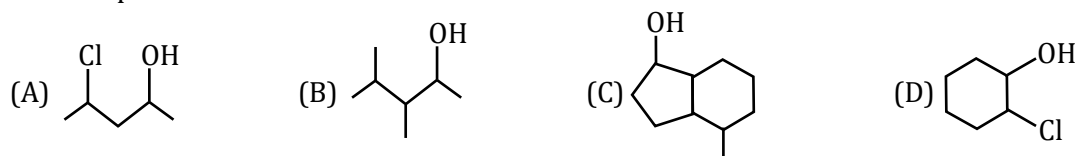
40. Number of chiral carbon atoms in the compound X, Y and Z respectively would be :



- (A) 2, 1, 1 (B) 1, 1, 1 (C) 2, 0, 2 (D) 2, 0, 0

COI102

41. The compound which has maximum number of chiral centres is



COI103

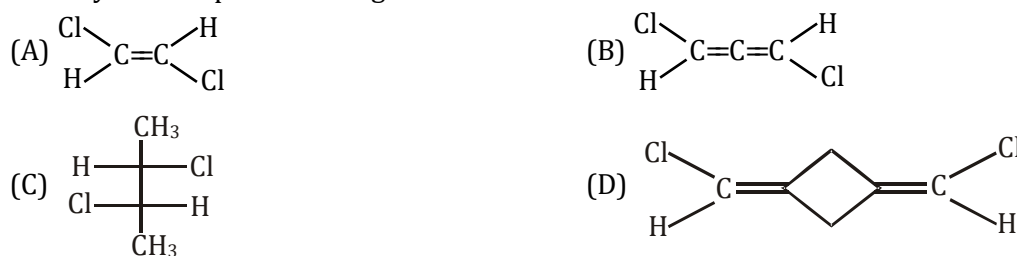
42. Which of the following amine is optically active.

- (A) CH_3NH_2 (B) $\text{CH}_3\text{NHC}_2\text{H}_5$ (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{-N}(\text{CH}_3)(\text{C}_2\text{H}_5)$ (D) Sec-Butylamine

COI121

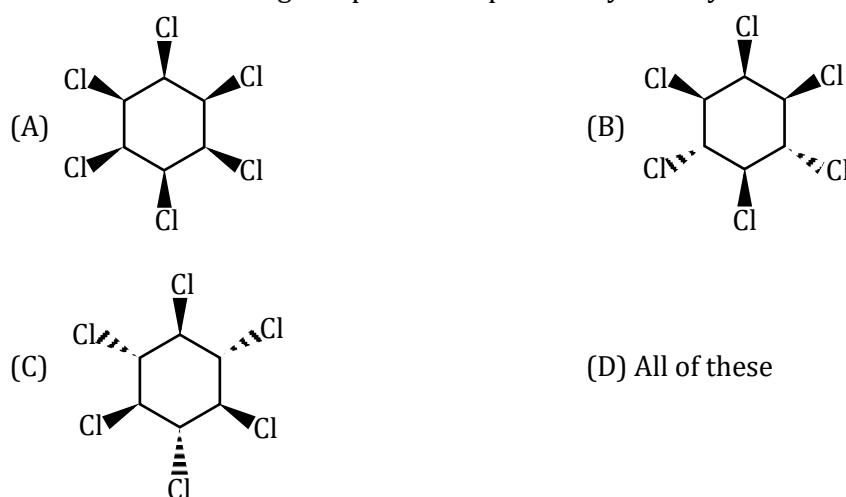
Symmetry Elements: Centre of Symmetry

43. Identify the compound having COS



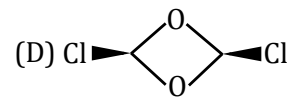
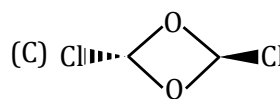
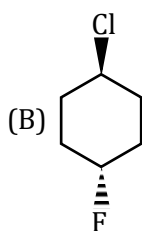
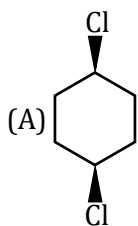
COI086

44. Which of the following compound has plane of symmetry and center of symmetry?



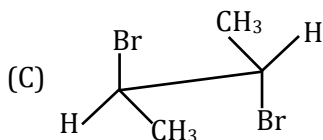
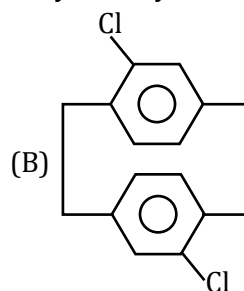
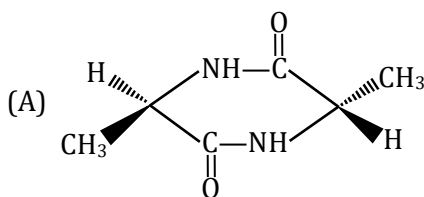
COI087

45. Compound having POS and COS both?



COI089

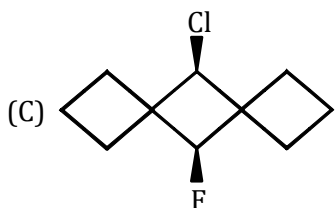
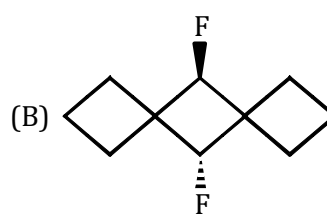
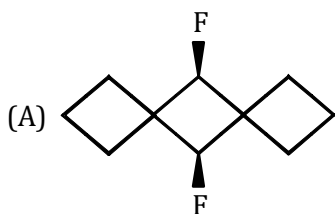
46. Which of the following compounds has centre of symmetry?



(D) All of these

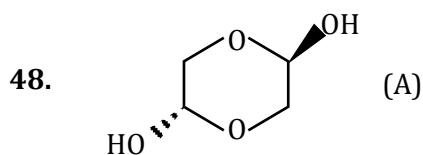
COI092

47. Which of the following containing both POS & COS?



(D) None of these

COI095



What is optical rotation of (A) ?

(A) $+50^\circ$

(B) -50°

(C) 0°

(D) 100°

COI120

Symmetry Elements: Axis of Symmetry; Alternate Axis of Symmetry

49. Which of the following has C_2 & C_3 axis of symmetry?

(A) NH_3

(B) CH_2Cl_2

(C) CH_3-Cl

(D) BCl_3

COI090

Symmetry Elements: Plane of Symmetry

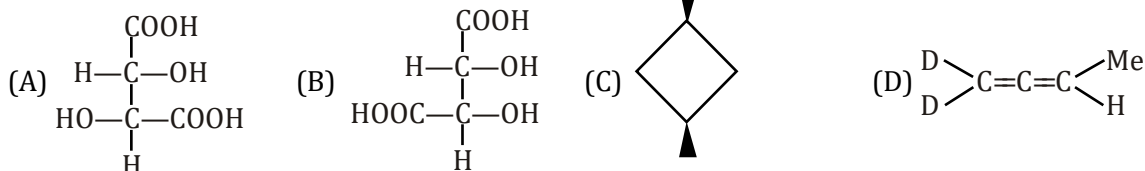
50. Number of POS present in benzene?

- (A) 5 (B) 6 (C) 7 (D) 9

COI091

Optical isomerism in allenes, biphenyls, spiro compounds and in rigid body system

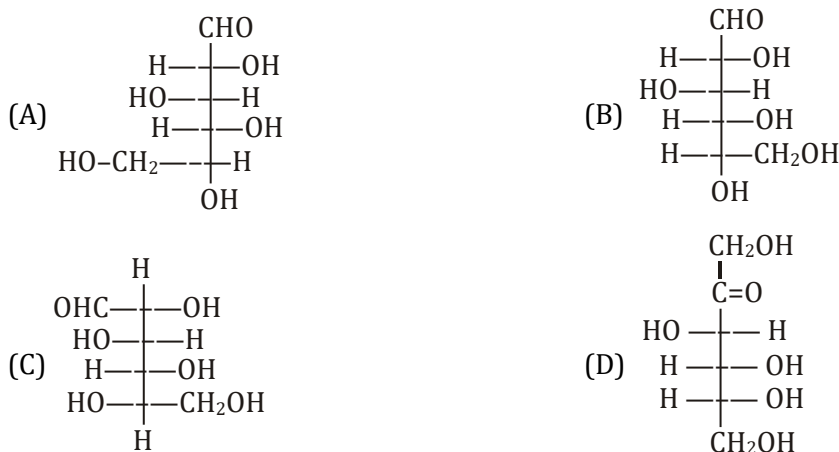
51. Identify optically active compound among following :



COI100

D, L Nomenclature

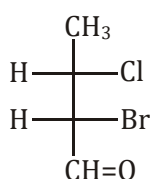
52. Which of the following is not D sugar :



COI109

R, S Nomenclature

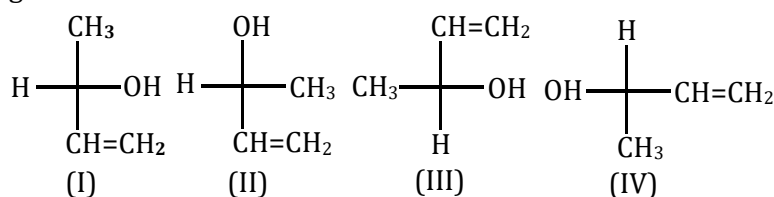
53. The correct nomenclature of following compound is :



- (A) 2R, 3R (B) 2S, 3R (C) 2R, 3S (D) 2S, 3S

COI106

54. Which of the following combinations amongst the four Fischer projections represents the same absolute configurations?



- (A) (II) and (III) (B) (I) and (IV) (C) (II) and (IV) (D) (III) and (IV)

COI107

Optical Purity and Enantiomeric Excess

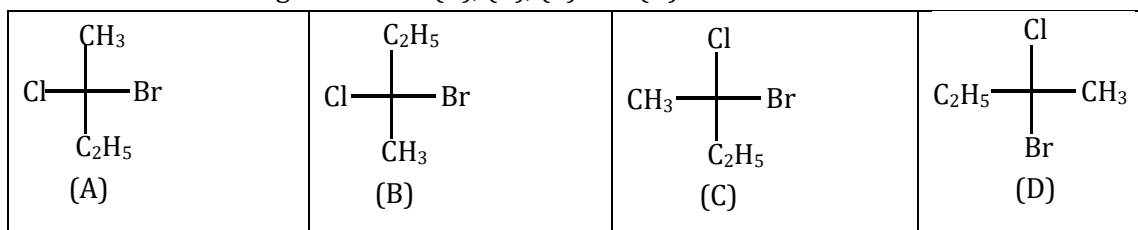
55. What is optical purity of $+9^\circ$ and -1° form ?

- (A) 40% (B) 80% (C) 10% (D) 60%

COI119

Assigning relationship in stereo isomers (Enantiomers, Diastereomers and identical isomers)

56. Consider the following structures (A), (B), (C) and (D) -

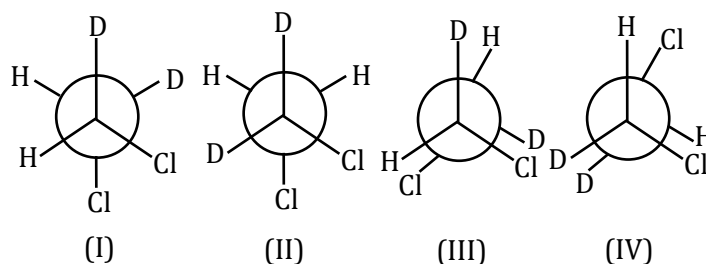


Which of the following statements is not correct

- (A) B and C are identical (B) A and B are enantiomers
(C) A and C are enantiomers (D) B and D are enantiomers

COI110

57. The identical compounds are



- (A) I, II (B) II, III (C) II, IV (D) I, III

COI112

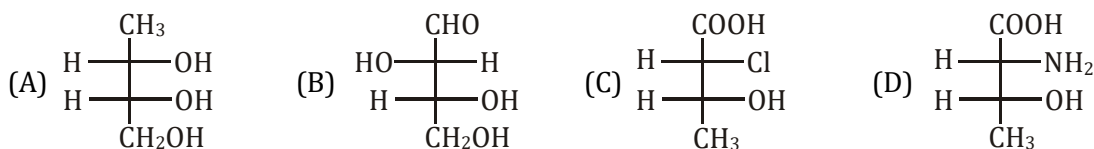
58. The total number of optically active isomer for $\text{CH}_2\text{OH}(\text{CHOH})_3\text{CHO}$ is -

- (A) 2 (B) 4 (C) 8 (D) 12

COI125

Erythro and Threo

59. Which of the following is a 'threo' isomer :



COI108

Calculation of Optical Isomers and total isomers

60. A compound can be divided into two equal halves & contains even 'n' asymmetric carbon atoms. The number of stereo isomer is -

- (A) 2^n (B) $2^{(n-1)}$ (C) $2^{\frac{n}{2}-1}$ (D) $2^{n-1} + 2^{\frac{n}{2}-1}$

COI122

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

STRUCTURAL ISOMERISM

Functional Isomerism

1. Compound having molecular formula $C_nH_{2n-4}O_3$ can have functional group.
 (A) 3-Aldehyde group (B) 1-Carboxylic acid & 2-Aldehyde
 (C) 1-Carboxylic acid anhydride & 1-Alcohol (D) 1-Carboxylic acid & 1-Alcohol

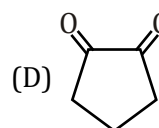
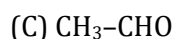
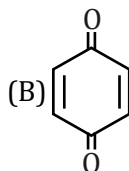
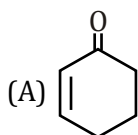
CS1105

Tautomerism and it's Mechanism

2. Correct statements regarding compounds having molecular formula $C_5H_{10}O$ is :
 (A) It has four structurally isomeric aldehyde, all can show tautomerism
 (B) It has four structurally isomeric aldehyde, out of four three can show tautomerism
 (C) It has three structurally isomeric ketones, all can show tautomerism
 (D) It has three structurally isomeric ketones, no one can show tautomerism

CS1106

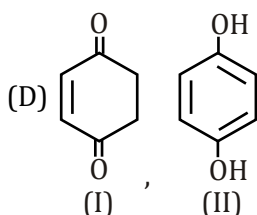
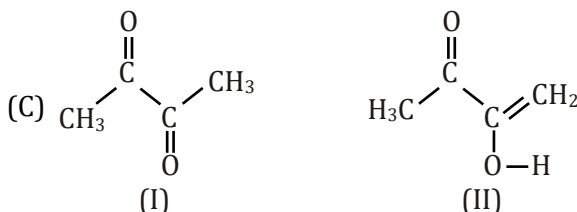
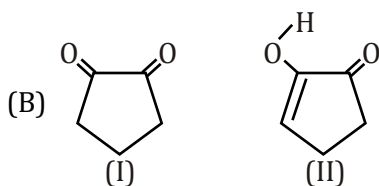
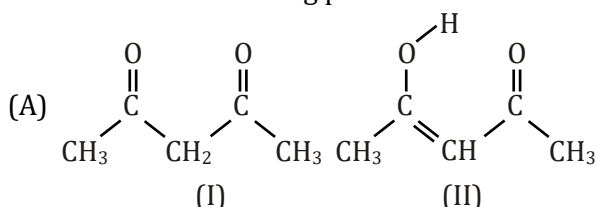
3. Which of the following can show tautomerism ?



CS1107

Enol Content and Kinetic Isotopic Effect

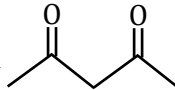
4. In which of the following pairs 2nd is more stable than its 1st form :

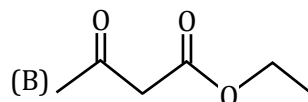
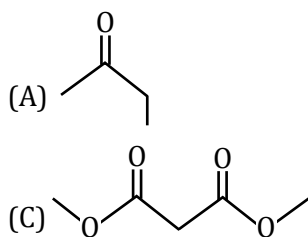


CS1108

5. Which of the following statements is/are not correct :
- (A) Metamerism belongs to the category of structural isomerism
 (B) Tautomeric structures are the resonating structures of a molecule
 (C) Keto form is always more stable than the enol form
 (D) None of these

CS1109

6. Which compound has less enol content w.r. to enol content of compound 



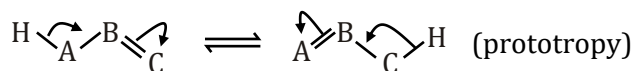
(D) None of these

CS1110

COMPREHENSION TYPE QUESTIONS

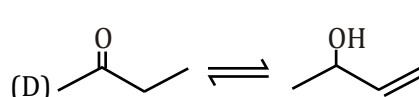
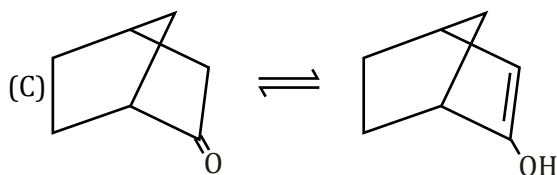
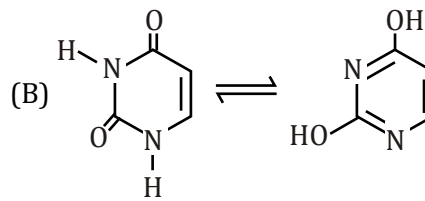
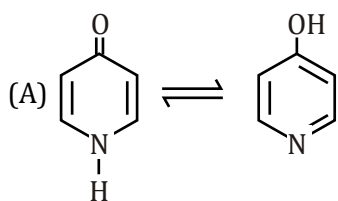
Paragraph (Q. 7 to Q.8)

Tautomerism involves interconversion of suitable functional isomers by the 1,3-shift of a hydrogen from an atom which is itself bonded to a functional group containing a double bond, eg : C=O, C=N, N=O and C=C to the terminal atom of the double bond.



Tautomerism and its Mechanism

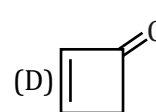
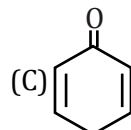
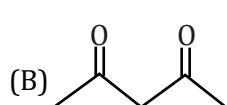
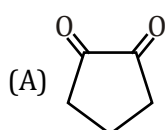
7. Which of the following pair are an example of tautomerism



CS1111

Enol Content and Kinetic Isotopic Effect

8. Which of the following having higher percentage of enol form :



CS1112

GEOMETRICAL ISOMERISM AND CONFORMATIONS

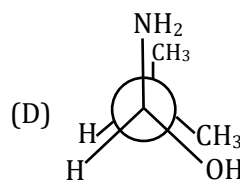
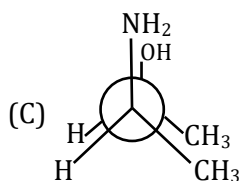
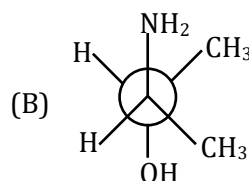
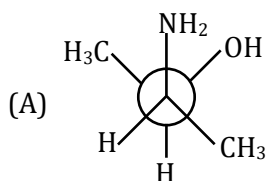
COMPREHENSION TYPE QUESTIONS

Questions on Stability and Dipole moment

Paragraph for Question 9 & 10

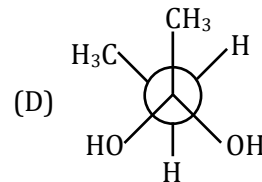
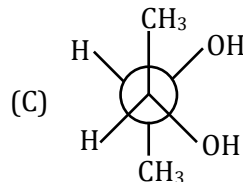
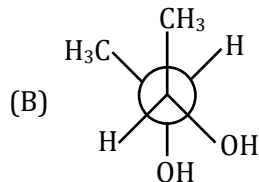
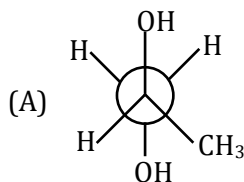
Different special arrangements of atoms that result from free rotation about a single bond are called conformation. Stability of conformer depends on torsional strain, steric strain, dipole-dipole attraction and intramolecular hydrogen bonding.

9. Which one of the following conformers of 3-amino-2-butanol is most stable:



CGI107

10. In which of the following has minimum potential energy in conformers of butane-2-3-diol :



CGI108

MULTIPLE CORRECT TYPE QUESTIONS

General Definition of Stereoisomerism, Conditions of Geometrical Isomerism in Alkene

11. For a compound with IUPAC name P_1 -methyl pent- P_2 -ene which is/are correctly matched. Where P_1 & P_2 are position of substituent & multiple bond respectively.
- (A) $P_1 = 2, P_2 = 2$ compound doesn't show geometrical isomerism
 - (B) $P_1 = 2, P_2 = 1$ compound doesn't show geometrical isomerism
 - (C) $P_1 = 2, P_2 = 3$ compound shows optical isomerism
 - (D) $P_1 = 3, P_2 = 2$ compound doesn't show geometrical isomerism

CGI140

Geometrical Isomerism in Amines, Oximes and Azo Compounds

12. Which of the following carbonyl compounds can give two oximes on reaction with hydroxyl amine?
- (A) HCHO (B) CH_3CHO (C) PhCHO (D) CH_3COPh

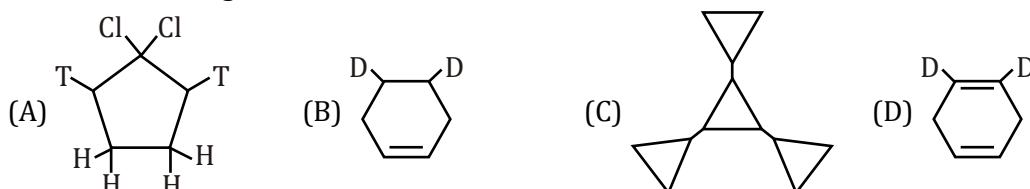
CGI141

Geometrical Isomerism in Cycloalkanes and Cycloalkenes

13. Which of the following is correct statement :
- (A) Geometrical isomers are not mirror image isomer.
 (B) A compound having double bond (restricted bond) always show geometrical isomerism.
 (C) Acyclic compound having only single bond does not show geometrical isomerism.
 (D) Cyclodecene can show cis & trans form.

CGI142

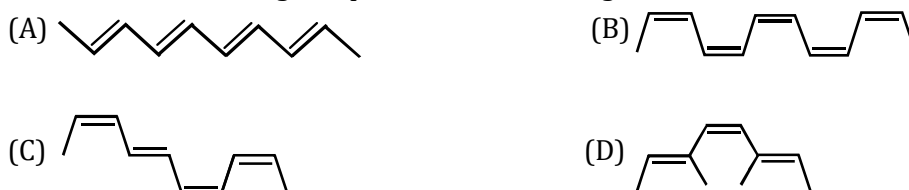
14. Which will show geometrical isomerism -



CGI143

Nomenclature of Geometrical Isomers, Cis-Trans Nomenclature, CIP Rules, E-Z Nomenclature and Syn-Anti Nomenclature

15. Which of the following compounds has cis configuration at each double bond?

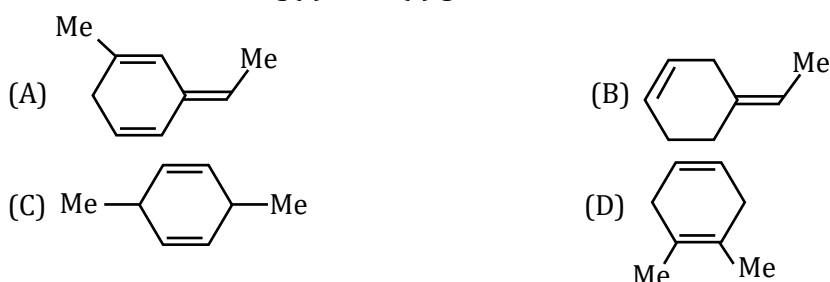


CGI144

16. Which of the following is true for maleic acid and fumaric acid.
- (A) Configurational isomers (B) Stereo isomers
 (C) Z and E isomers (D) Constitutional isomers

CGI145

17. Which of the following(s) show(s) geometrical isomerism?

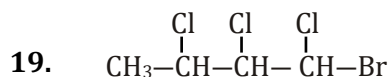


CGI146

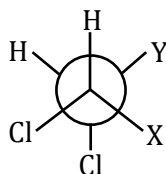
Questions on Stability and Dipole moment

18. For a compound 1, 2-dichloro ethane which of the following statement are correct.
- (A) Anti-conformation is most stable one
 (B) A Gauche conformation has non-zero dipole moment
 (C) Eclipsed conformation has plane of symmetry
 (D) None of these

CGI147



For the above compound (A) the given newmann projection can be represented



X & Y are respectively:

(A) CH_3 and CHClBr

(B) CH_3 & CH_3

(C) $-\underset{\text{Cl}}{\text{CH}}-\text{CH}_3$ & Br

(D) $-\underset{\text{CH}_3}{\text{CH}}-\text{Cl}$ & $-\text{CH}_3$

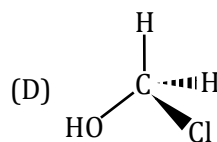
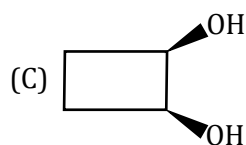
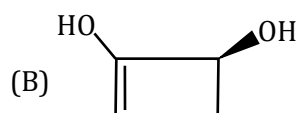
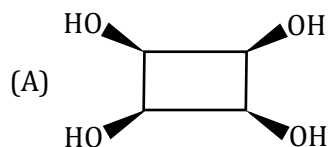
CGI148

OPTICAL ISOMERISM

MULTIPLE CORRECT TYPE QUESTIONS

Symmetry Elements: Plane of Symmetry

20. Which of the following compound have plane of symmetry?



COI097

Symmetry Elements: Axis of Symmetry; Alternate Axis of Symmetry

21. The number of molecule that posses C_2 -axis of symmetry?

(A) BF_3

(B) Benzene

(C) cis-2-butene

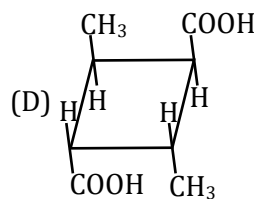
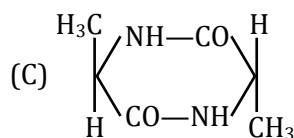
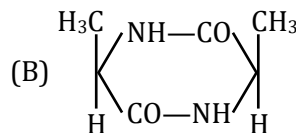
(D) cyclopropane

COI096

Optical Resolution

22. Which of the following are not resolvable ?

(A) 2,3-Pentadiene



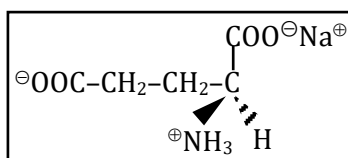
COI126

COMPREHENSION TYPE QUESTIONS

Optical Purity and Enantiomeric Excess

Paragraph for Question 23 to 25

S(+) Mono sodium Glutamate (MSG) is a flavour enhancer used in many foods. Fast foods often contain substantial amount of MSG and is widely used in Chinese food. If one mole of above MSG was placed in 845 ml solution and passed through 200 mm tube, the observed rotation was found to be $+9.6^\circ$.



23. Find out the specific rotation of (-) MSG :
 (A) $+24^\circ$ (B) $+56.8^\circ$ (C) -48° (D) None of these
24. Find out the approximate percentage composition of (-) MSG in a mixture containing (+) MSG and (-) MSG whose specific optical rotation is -20° :
 (A) 83.3% (B) 16.7% (C) 91.6% (D) 74%
25. If 33.8 g of (+) MSG was put in 338 ml solution and was mixed with 16.9g of (-) MSG put in 169 ml solution and the final solution was passed through 400 mm tube. Find out observed rotation of the final solution:
 (A) $+1.6^\circ$ (B) $+4.8^\circ$ (C) $+3.2^\circ$ (D) None of these

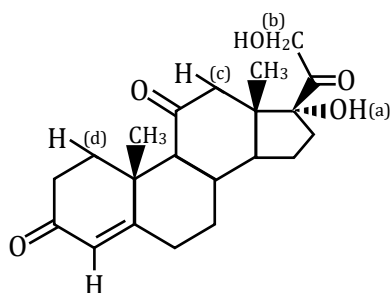
COI083

COI084

COI085

Calculation of Optical Isomers and total isomers

Paragraph for Question No. 26 to 27



Cortisone

26. Total number of chiral centers in Cortisone :-
 (A) 4 (B) 5 (C) 6 (D) 7
27. Total number of possible stereoisomers of the compound Cortisone will be :
 (A) 32 (B) 64 (C) 66 (D) 128

COI116

COI117

EXERCISE – (S)

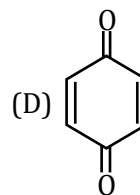
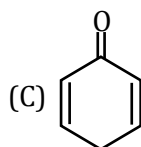
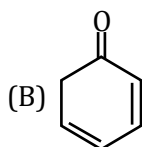
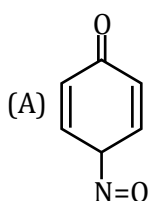
Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

STRUCTURAL ISOMERISM

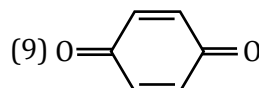
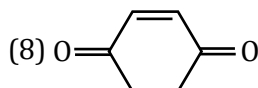
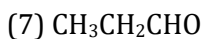
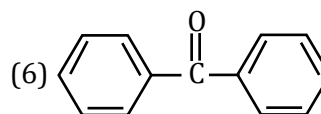
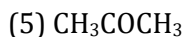
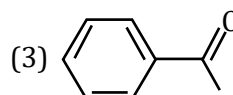
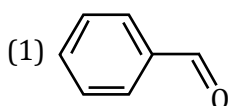
Tautomerism and its Mechanism

1. Number of compounds which undergoes Tautomerisation to form aromatic product are :



CS1113

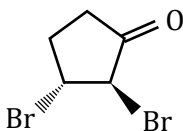
2. Among the following how many of them will show keto enol tautomerism ?



CS1085

Enol Content and Kinetic Isotopic Effect

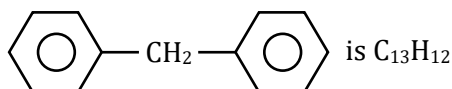
3. How many hydrogen in the compound below are replaced by deuterium when it is shaken in D_2O containing trace of hydroxide ion?



CS1083

Identification and Number of Structural Isomers

4. The molecular formula of diphenyl methane



How many structural isomers are possible when one of the hydrogen is replaced by a chlorine atom?

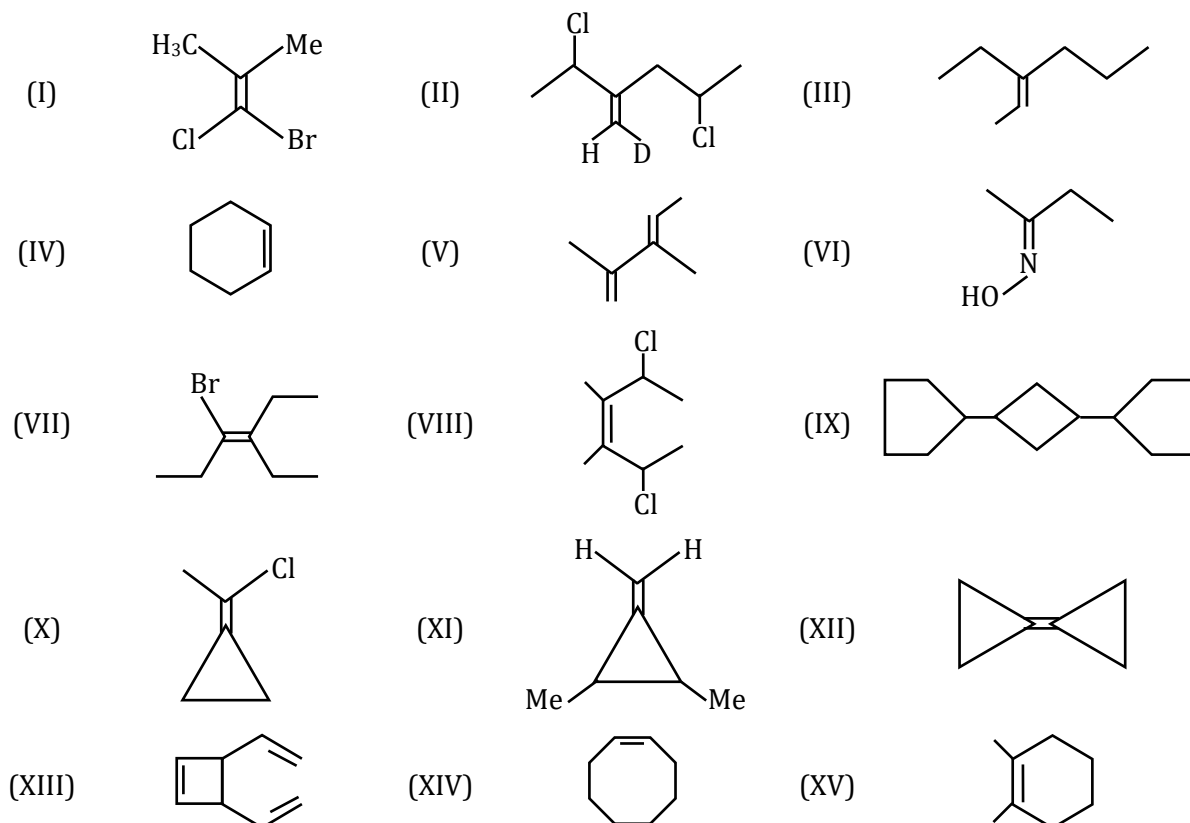
CS1077

5. How many 5 membered parent chain alkane are possible for C_7H_{16} ? CS1078
6. Total number of possible structural isomers of $C_5H_{11}Br$. CS1079
7. How many structural isomer are possible for $C_4H_{10}O$ (only alcohol). CS1081
8. Number of structural isomer of C_6H_{14} is. CS1082
9. How many total number of structural isomer are possible for C_4H_7Cl having parent chain of four carbon. CS1022
10. How many structural isomers will be possible for molecular formula C_5H_{10} (only cyclic) where cyclopropane ring is parent carbon system? CS1025
11. Total number of esters and acids possible for the compound having molecular C_4H_8O (only structural)? CS1027

GEOMETRICAL ISOMERISM AND CONFORMATIONS

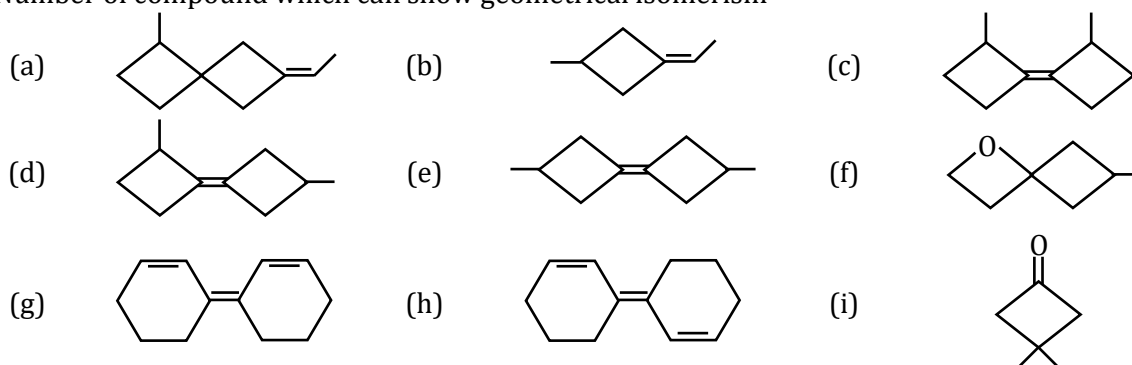
Geometrical Isomerism in Cycloalkanes and Cycloalkenes

12. Identify molecules which usually can show geometrical isomerism (at room temperature)



CG1086

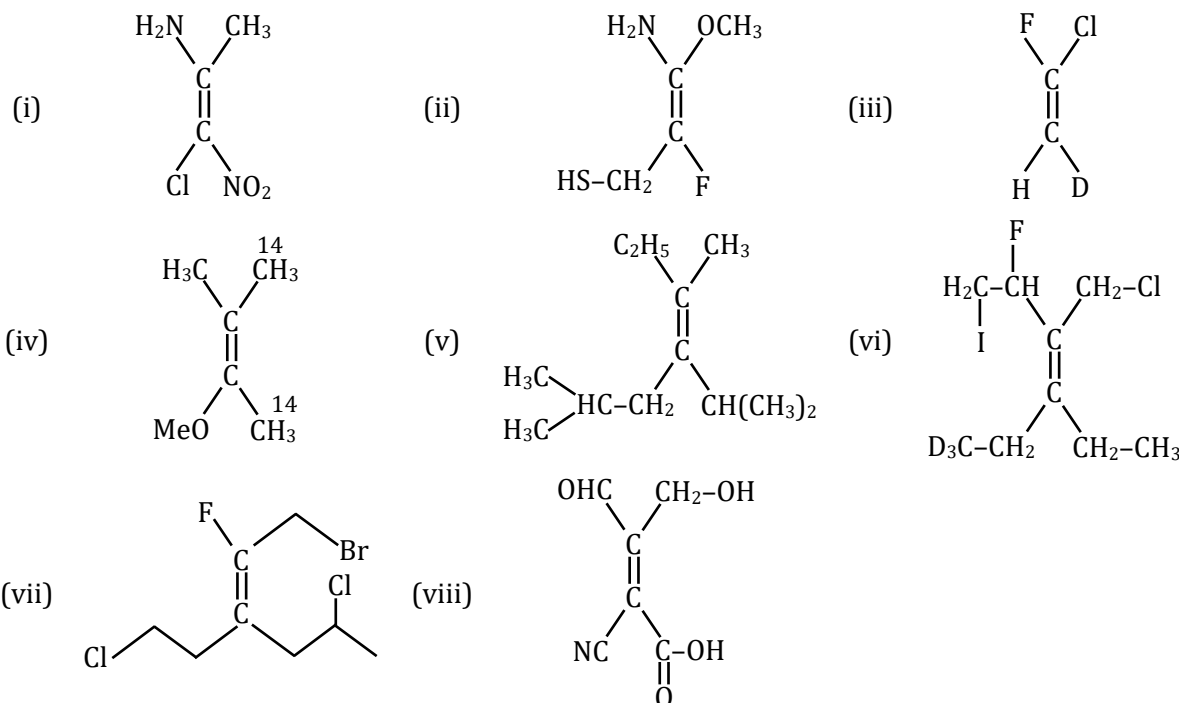
13. Number of compound which can show geometrical isomerism



CG1024

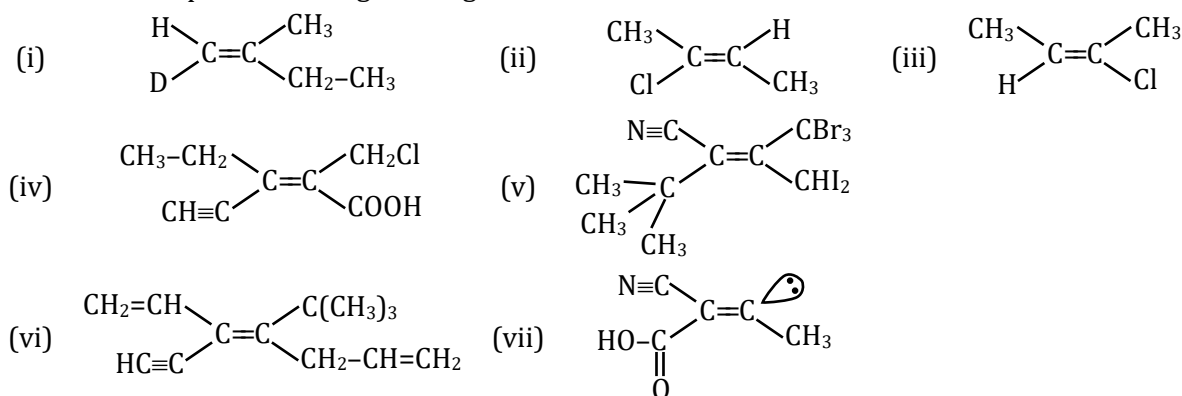
Nomenclature of Geometrical Isomers, Cis-Trans Nomenclature, CIP Rules, E-Z Nomenclature and Syn-Anti Nomenclature

14. How many of the following have E configurations:

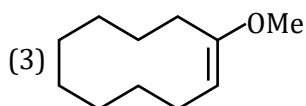
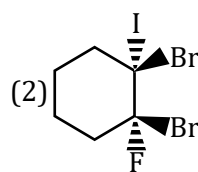
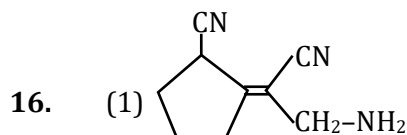


CG1102

15. Number of compounds having z-configuration:



CG1103



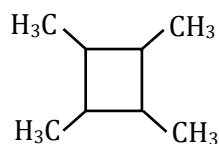
Assign E or Z to the following compounds and write 1 for E and 2 for Z.

Write answer of part (1), (2), (3) & (4) in the same order and present the four digit number as answer in OMR sheet. For example: If all these answer are 9 then fill 9999 in OMR sheet.

CGI022

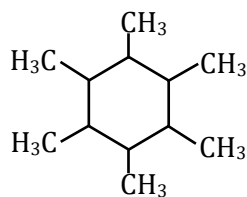
Pseudo Geometrical Isomerism Unit and Calculation of Total Geometrical Isomers

17. How many geometrical isomers are possible for the following compound?



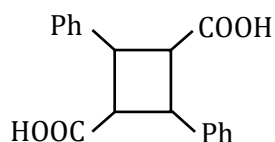
CGI135

18. How many geometrical isomers are possible for the following compound?



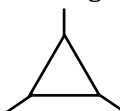
CGI136

19. How many geometrical isomers are possible for the following compound?



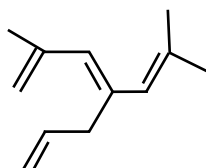
CGI137

20. Number of geometrical isomer for the following compound will be



CGI030

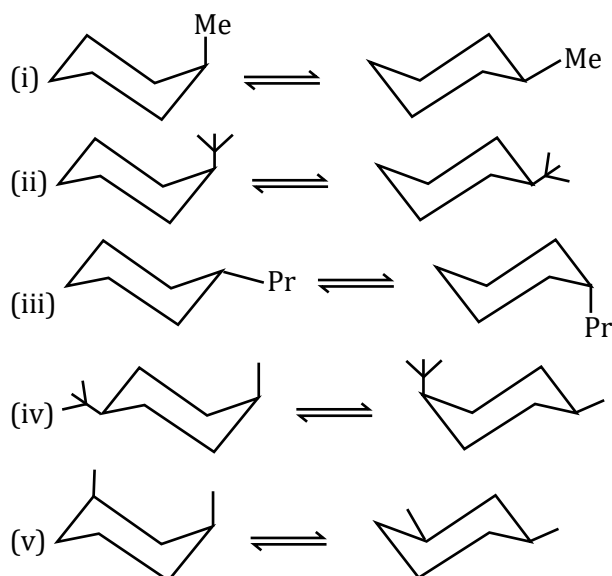
21. Number of geometrical isomers for following compound.



CGI025

Axial-Equatorial orientation in Cyclohexane, Ring Flipping; Keq of Flipping; Geometrical Isomerism in Cyclohexane and 1;3 Diaxial Repulsion

22. The ring flipping equilibrium constant is greater than 1 for

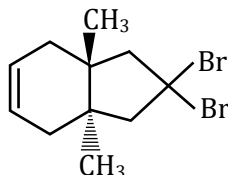


CGI027

OPTICAL ISOMERISM

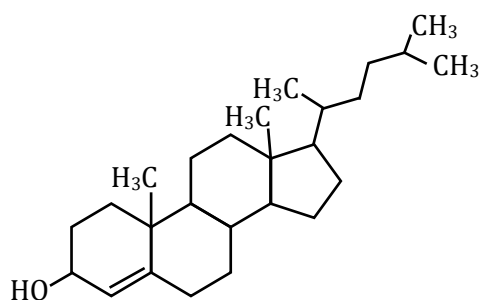
Chiral Compound and Chiral Centre and asymmetric centre and pseudo chiral centre

23. How many chiral centres are present in the following compound?



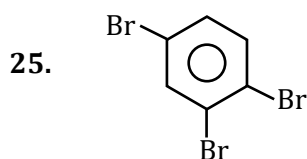
COI141

24. How many chiral centres are present in the following compound?



COI142

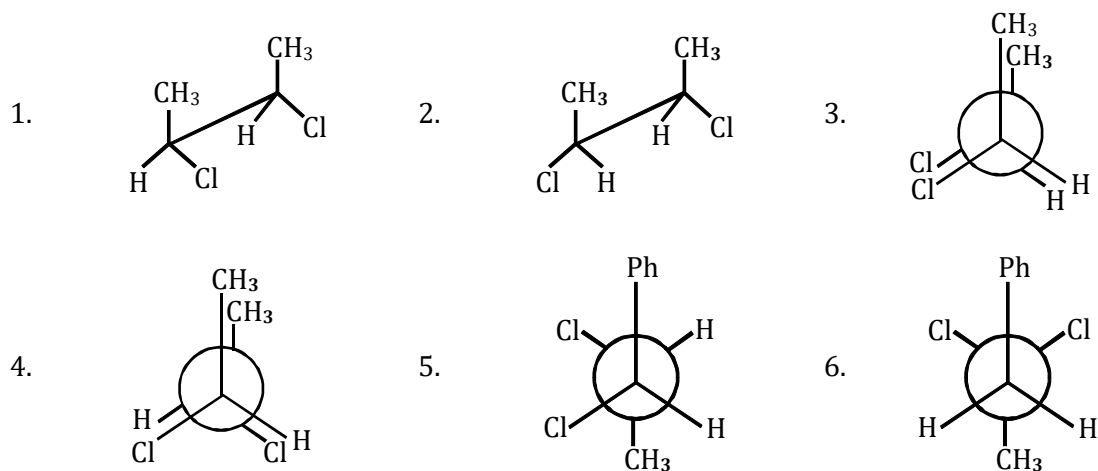
Symmetry Elements: Plane of Symmetry



Number of plane of symmetry in above compound is :

COI098

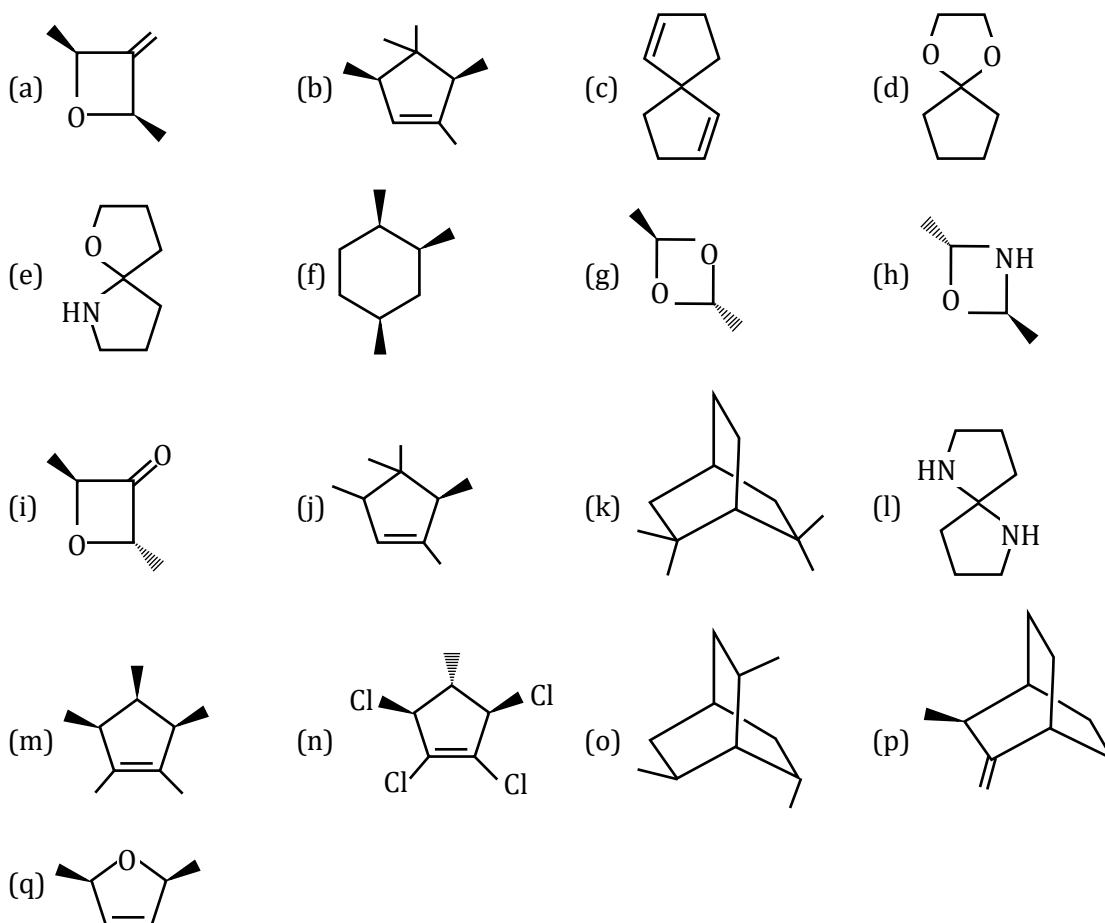
26. Identify the no of compounds containing POS.



COI099

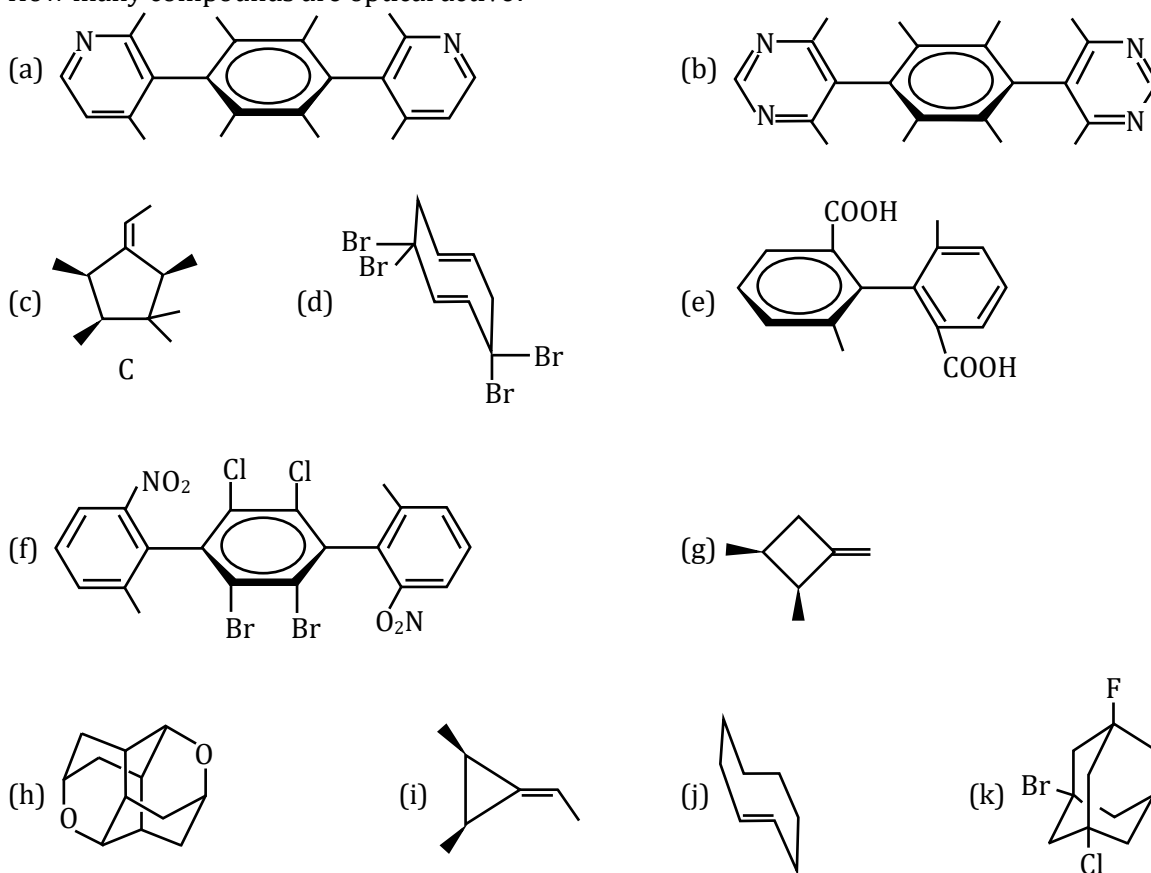
Optical isomerism in allenes, biphenyls, spiro compounds and in rigid body system

27. How many compounds are optically inactive?



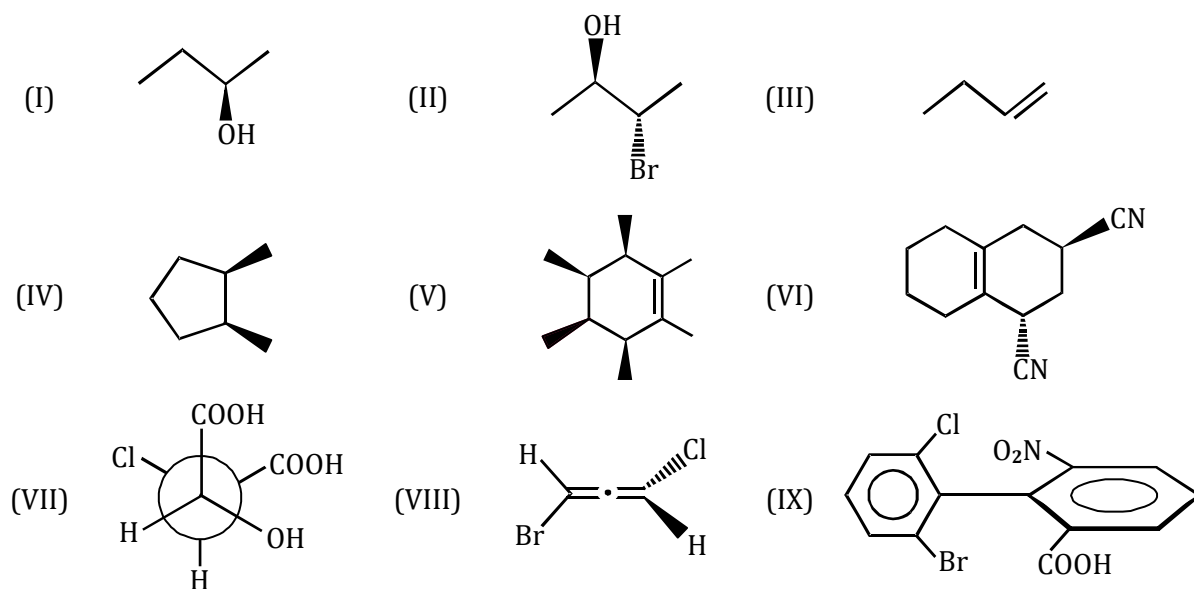
COI132

28. How many compounds are optical active?



COI133

29. How many of the given compounds are chiral:



COI026

Optical Purity and Enantiomeric Excess

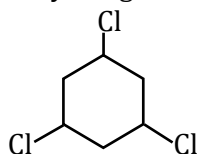
30. A mixture having 47% d-tartaric acid & 53% l-tartaric acid

What is the optical purity of the mixture w.r.t. l-isomer :

COI028

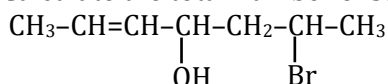
Calculation of Optical Isomers and total isomers

31. Total number of stereoisomers formed by the given compound is:



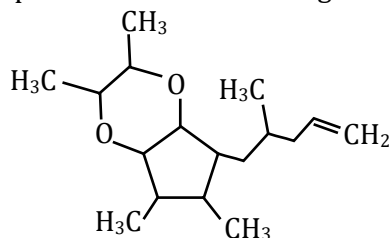
COI130

32. Calculate the total number of stereoisomers for the given compound:



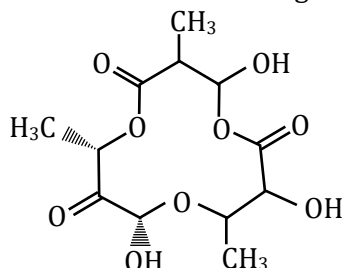
COI131

33. How many stereoisomers are possible for the following compound?



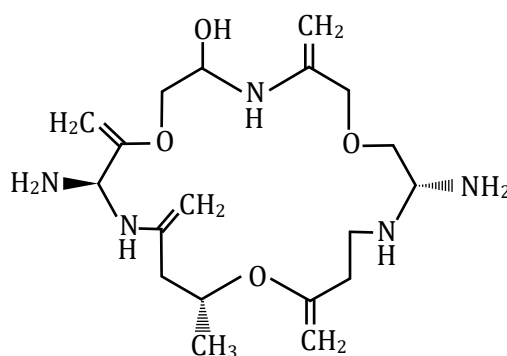
COI134

34. How many stereoisomers are possible for the following compound?



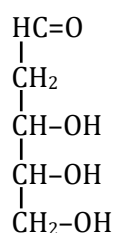
COI135

35. How many stereoisomers are possible for the following compound?



COI137

36. For the given aldopentose how many stereoisomers are possible.



COI029

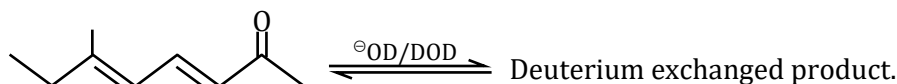
Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

STRUCTURAL ISOMERISM

Enol Content and Kinetic Isotopic Effect

37. Number of H(Hydrogens) will be replaced by D (Deuterium) in given compound when it is kept in mild basic medium for a long time is Z then $\frac{Z}{3}$ is



CS1116

Identification and Number of Structural Isomers

38. Total number of structural isomers of C_9H_{18} containing cyclohexane ring is x then $(x-1.5)$
39. The number of Benzenoid isomers possible for $\text{C}_6\text{H}_3\text{ClBrI}$ is x then $\left(\frac{x}{3}\right)$ is

CS1114

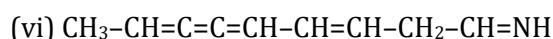
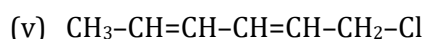
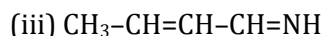
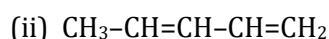
CS1115

GEOMETRICAL ISOMERISM AND CONFORMATIONS

Pseudo Geometrical Isomerism Unit and Calculation of Total Geometrical Isomers

40. Total number of G.I. in following compounds is x then $\frac{x}{2}$.

Write answer as the sum of all the values obtained :



CG1149

Questions on Stability and Dipole moment

41. $\mu_{\text{obs}} = \sum \mu_i x_i$; where μ_i is the dipole moment of stable conformer and x_i is the mole fraction of that conformer of $\text{Z-CH}_2\text{-CH}_2\text{-Z}$ in Newmann's projection. If μ_{solution} or $\mu_{\text{net}} = 1 \text{ D}$ and mole fraction of anti form = 0.82, find μ_{Gauche} .

CG1138

OPTICAL ISOMERISM

Calculation of Optical Isomers and total isomers

42. X = total number of isomers of C_7H_{14} (only 5-membered ring).

Y = number of optically active isomers of C_7H_{14} (only 5-membered ring).

Then $\frac{Y}{X} = ?$

COI148

43. Number of pair of enantiomers possible for 1-bromo- 2-chlorocyclohexane if X .

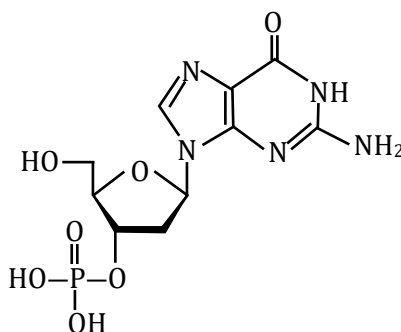
Number of pair of enantiomers possible for 1-chloro- 2-fluorocyclobutane if Y .

Then find out the value of $X + Y$:

COI030

44. X = stereoisomers possible for the following compound,

Y = Number of chiral centers. Then $\frac{Y}{X} = ?$



COI149

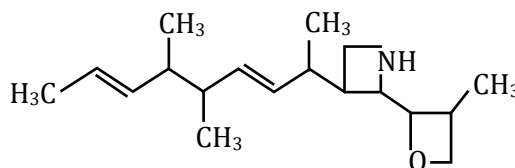
45. X = Total number of isomers for $C_4H_6Br_2$ containing cyclobutane ring (including stereoisomer),
 Y = number of optically active isomers for $C_4H_6Br_2$ containing cyclobutane ring (including stereoisomer)

Then $\frac{X}{Y} = ?$

COI150

46. If X = stereoisomers are possible for the following compound,

Then $\frac{X}{100} = ?$



COI151

EXERCISE - JEE (Main) PYQ

STRUCTURAL ISOMERISM

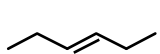
1. Which of the following pairs of compounds are positional isomers:-

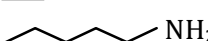
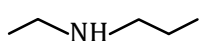
[JEE (Main) 2015]

- (1) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-CH}_3$ and $\text{CH}_3\text{-CH}_2\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-CH}_2\text{-CH}_3$
 (2) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-CH}_3$ and $\text{CH}_3\text{-}\overset{\text{CH}_3}{\underset{|}{\text{CH}}}\text{-CH}_2\text{-CHO}$
 (3) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CHO}$ and $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-CHO}$
 (4) $\text{CH}_3\text{-CH}_2\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-CH}_2\text{-CH}_3$ and $\text{CH}_3\text{>CH-CH}_2\text{-CHO}$

CS1036

2. Given below are two statements :

Statement (I) :  and  are isomeric compounds.

Statement (II) :  and  are functional group isomers.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) Both **Statement I** and **Statement II** are false
 (2) Both **Statement I** and **Statement II** are true
 (3) **Statement I** is true but **Statement II** is false
 (4) **Statement I** is false but **Statement II** is true

CS1122

3. Compound with molecular formula $\text{C}_3\text{H}_6\text{O}$ can show:

[JEE (Main) 2021]

- (1) Positional isomerism
 (2) Both positional isomerism and metamerism
 (3) Metamerism
 (4) Functional group isomerism

CS1087

4. Monocarboxylic acids are functional isomers of :

[JEE (Main) 2013]

- (1) Esters (2) Amines (3) Ethers (4) Alcohols

CS1035

5. Match **List - I** with **List - II**.

[JEE (Main) 2024]

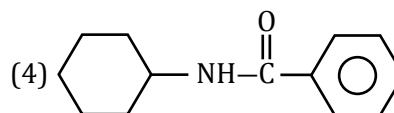
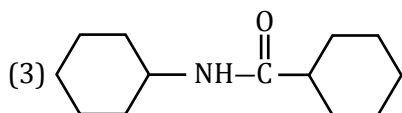
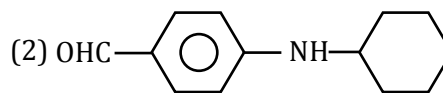
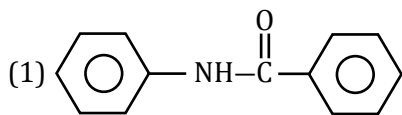
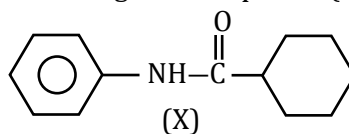
List- I (Pair of Compounds)		List- II (Isomerism)	
A.	n-propanol and Isopropanol	I.	Metamerism
B.	Methoxypropane and ethoxyethane	II.	Chain Isomerism
C.	Propanone and propanal	III.	Position Isomerism
D.	Neopentane and Isopentane	IV.	Functional Isomerism

- (1) (A)–(II), (B)–(I), (C)–(IV), (D)–(III) (2) (A)–(III), (B)–(I), (C)–(II), (D)–(IV)
 (3) (A)–(I), (B)–(III), (C)–(IV), (D)–(II) (4) (A)–(III), (B)–(I), (C)–(IV), (D)–(II)

CS1117

6. Which of the following is metamer of the given compound (X)?

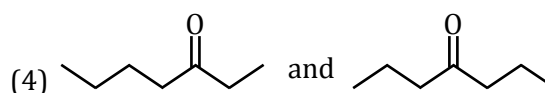
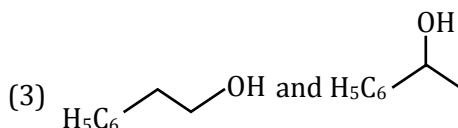
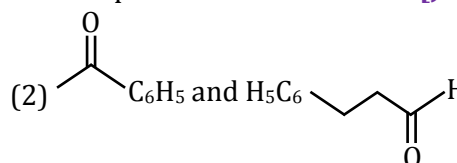
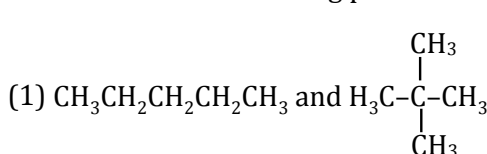
[JEE (Main) 2024]



CS1118

7. Which one of the following pairs of isomers is an example of metamerism?

[JEE (Main) 2021]



CS1088

8. Match List-I with List-II.

[JEE (Main) 2023]

List- I Isomeric pairs		List- II Type of isomers	
A.	Propanamine and N-Methylethanamine	I.	Metamers
B.	Hexan-2-one and Hexan-3-one	II.	Positional isomers
C.	Ethanamide and Hydroxyethanimine	III.	Functional isomers
D.	o-nitrophenol and p-nitrophenol	IV.	Tautomers

Choose the correct answer from the options given below :-

(1) A-III, B-IV, C-I, D-II

(2) A-IV, B-III, C-I, D-II

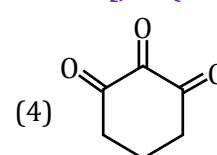
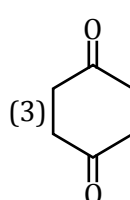
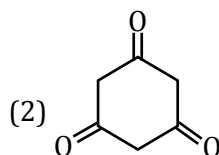
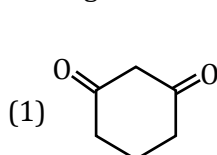
(3) A-II, B-III, C-I, D-IV

(4) A-III, B-I, C-IV, D-II

CS1039

9. Highest enol content will be shown by:

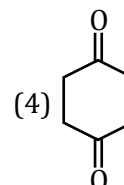
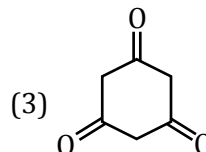
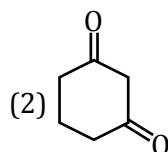
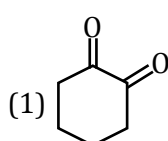
[JEE (Main) 2024]



CS1119

10. Which will have the highest enol content?

[JEE (Main) 2022]

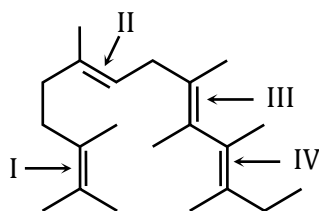


CS1089

11. Assertion A : Enol form of acetone [CH_3COCH_3] exists in $< 0.1\%$ quantity. However, the enol form of acetyl acetone [$\text{CH}_3\text{COCH}_2\text{OCCH}_3$] exists in approximately 15% quantity.
Reason R : enol form of acetyl acetone is stabilized by intramolecular hydrogen bonding, which is not possible in enol form of acetone.
Choose the correct statement : [JEE (Main) 2021]
(1) A is false but R is true
(2) Both A and R are true and R is the correct explanation of A
(3) Both A and R are true but R is not the correct explanation of A
(4) A is true but R is false CS1086
12. The number of different chain isomers for C_7H_{16} is _____. [JEE (Main) 2024]
CS1120
13. The total number of monobromo derivatives formed by the alkanes with molecular formula C_5H_{12} is (excluding stereo isomers) _____. [JEE (Main) 2022]
CS1038
14. The number of structural isomers for C_6H_{14} is:
(1) 6 (2) 4 (3) 3 (4) 5 [JEE (Main) 2015]
CS1037
15. How many cyclic structures are possible for C_4H_6 :-
(1) 3 (2) 5 (3) 4 (4) 6 [JEE (Main) 2012]
CS1034

GEOMETRICAL ISOMERISM AND CONFORMATIONS

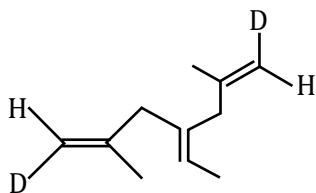
16. In the following structure, the double bonds are marked as I, II, III and IV



Geometrical isomerism is **not** possible at site(s)

- (1) III (2) I (3) III and IV (4) I and III [JEE (Main) 2016]
CG1037
17. Which of the following compounds will exhibit geometrical isomerism? [JEE (Main) 2015]
(1) 2-Phenyl-1-butene (2) 1,1-Diphenyl-1-propene
(3) 1-Phenyl-2-butene (4) 3-Phenyl-1-butene CG1036
18. Maleic acid and fumaric acids are :- [AIEEE - 2012]
(1) Tautomers (2) Chain isomers
(3) Geometrical isomers (4) Functional isomers CG1034
19. The **incorrect** statements regarding geometrical isomerism are :
(A) Propene shows geometrical isomerism.
(B) Trans isomer has identical atoms/groups on the opposite sides of the double bond.
(C) Cis-but-2-ene has higher dipole moment than trans-but-2-ene.
(D) 2-methylbut-2-ene shows two geometrical isomers.
(E) Trans-isomer has lower melting point than cis isomer.
Choose the **CORRECT** answer from the options given below : [JEE (Main) 2025]
(1) (A), (D) and (E) only (2) (C), (D) and (E) only
(3) (B) and (C) only (4) (A) and (E) only CG1152

20. Number of geometrical isomers possible for the given structure is/are _____.



[JEE (Main) 2024]

CGI150

21. The number of acyclic structural isomers (including geometrical isomers) for pentene are _____

[JEE (Main) 2021]

CGI039

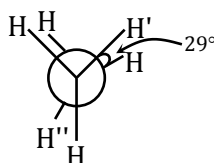
22. The dihedral angle in staggered form of Newman projection of 1, 1, 1-Trichloro ethane is degree. (Round off to the nearest integer)

[JEE (Main) 2021]

CGI040

23. In the following skew conformation of ethane, $H'-C-C-H''$ dihedral angle is :

[JEE (Main) 2019]

(1) 120° (2) 58° (3) 149° (4) 151°

CGI038

24. The incorrect statement regarding conformations of ethane is:

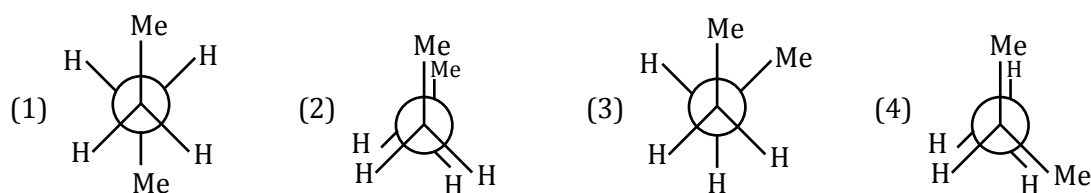
[JEE (Main) 2024]

- (1) Ethane has infinite number of conformations
- (2) The dihedral angle in staggered conformation is 60°
- (3) Eclipsed conformation is the most stable conformation.
- (4) The conformations of ethane are inter-convertible to one-another.

CGI151

25. Which of the following conformations will be the most stable?

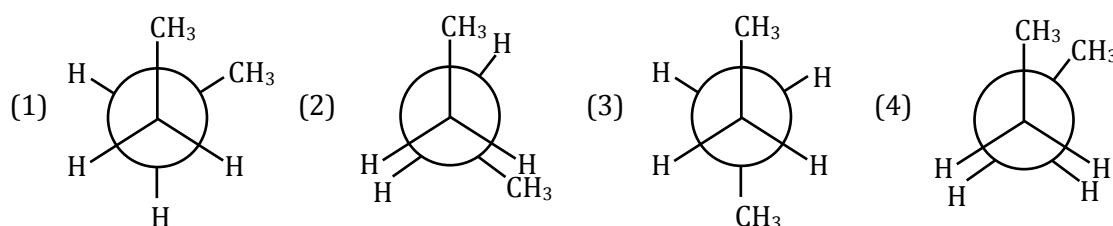
[JEE (Main) 2023]



CGI042

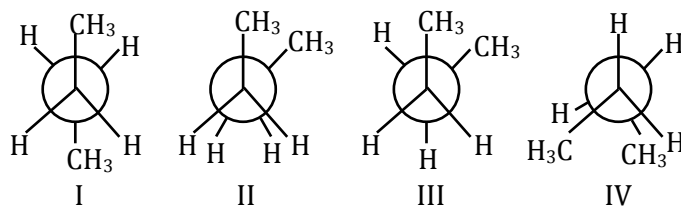
26. In the following structures, which one is having staggered conformation with maximum dihedral angle?

[JEE (Main) 2022]



CGI139

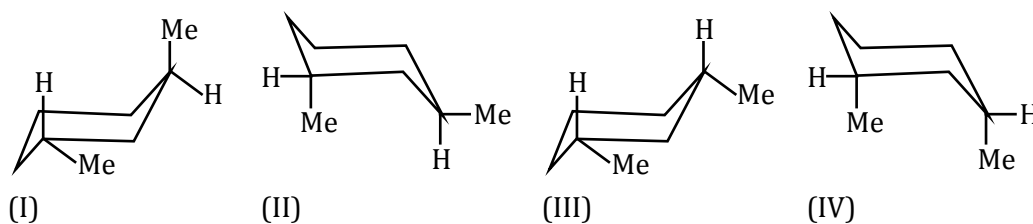
27. Arrange the following conformational isomers of n-butane in order of their increasing potential energy : [JEE (Main) 2021]



- (1) II < III < IV < I (2) I < IV < III < II (3) II < IV < III < I (4) I < III < IV < II

CGI041

28. Arrange in the correct order of stability (decreasing order) for the following molecules:



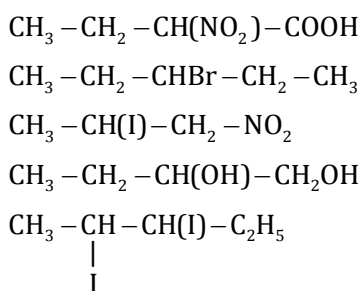
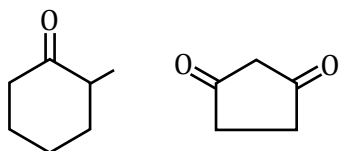
- (1) (I) > (II) > (III) > (IV) (2) (IV) > (III) > (II) ≈ (I)
(3) (III) > (I) ≈ (II) > (IV) (4) (I) > (II) ≈ (III) > (IV)

[JEE (Main) 2023]

CGI035

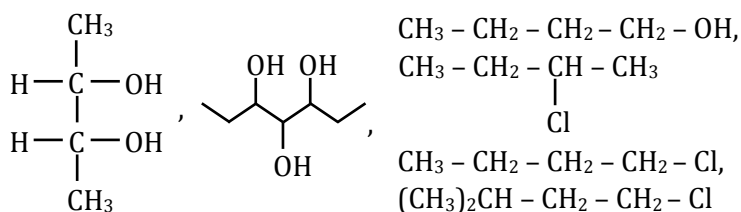
OPTICAL ISOMERISM

29. Total number of compounds with Chiral carbon atoms from following is _____. [JEE (Main) 2024]



COI152

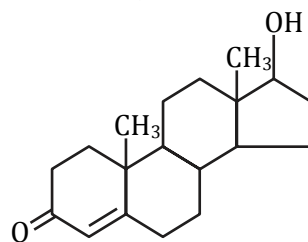
30. Total number of optically active compounds from the following is _____. [JEE (Main) 2024]



COI153

31. Testosterone, which is a steroidal hormone, has the following structure.

[JEE (Main) 2023]



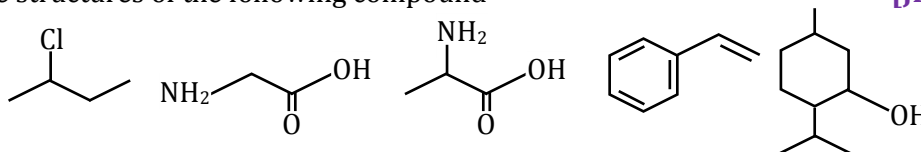
Testosterone

The total number of asymmetric carbon atom/s in testosterone is ____

COI147

32. Observe structures of the following compound

[JEE (Main) 2022]



The total number of structures/compounds which possess asymmetric carbon atoms is ____.

COI046

33. The optically inactive compound from the following is

[JEE (Main) 2015]

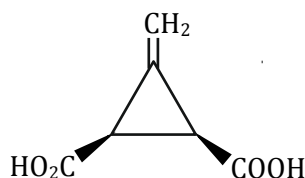
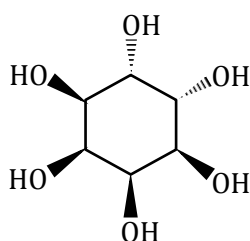
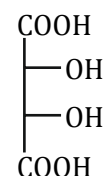
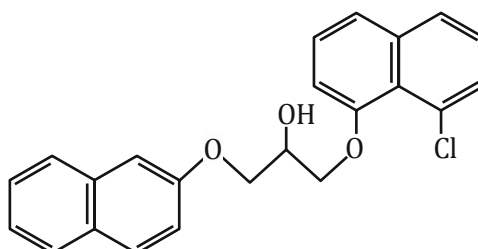
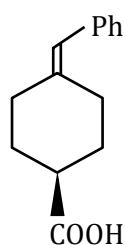
- (1) 2-chloropropanal (2) 2-chlorobutane
(3) 2-chloro-2-methylbutane (4) 2-chloropentane

34. Out of the following, the alkene that exhibits optical isomerism is :-

COI043
[AIEEE 2010]

- (1) 2-methyl-2-pentene (2) 3-methyl-2-pentene
(3) 4-methyl-1-pentene (4) 3-methyl-1-pentene

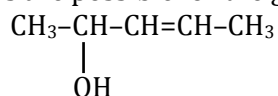
35. The total number of chiral compound/s from the following is ____.

COI041
[JEE (Main) 2023]

COI146

36. How many different stereoisomers are possible for the given molecule?

[JEE (Main) 2025]



(1) 3

(2) 1

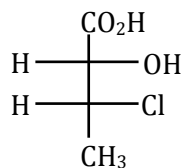
(3) 2

(4) 4

COI161

37. The absolute configuration of:

[JEE (Main) 2016]



(1) (2R, 3R)

(2) (2R, 3S)

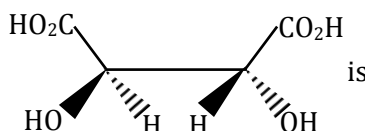
(3) (2S, 3R)

(4) (2S, 3S)

COI044

38. The absolute configuration of

[AIEEE - 2008]



(1) S, S

(2) R, R

(3) R, S

(4) S, R

COI039

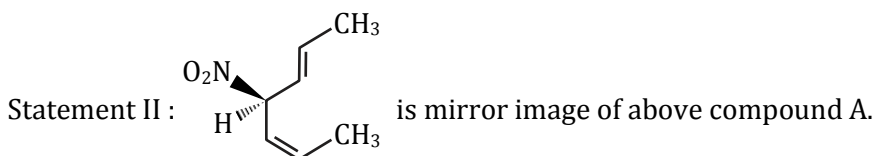
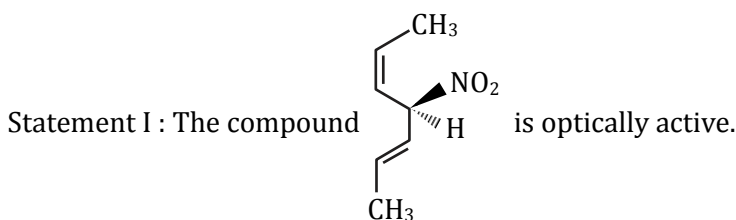
39. Optical activity of an enantiomeric mixture is $+12.6^\circ$ and the specific rotation of (+) isomer is $+30^\circ$. The optical purity is ____%

[JEE (Main) 2022]

COI143

40. Given below are two statements.

[JEE (Main) 2022]



In the light of the above statement, choose the **most appropriate** answer from the options given below.

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

COI144

41. The possible number of stereoisomers for 5-phenylpent-4-en-2-ol is _____.

[JEE (Main) 2025]

COI162

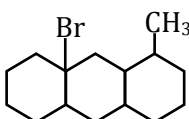
42. Number of optical isomers possible for 2-chlorobutane

[JEE (Main) 2024]

COI154

43. The number of optical isomers in following compound is : ____.

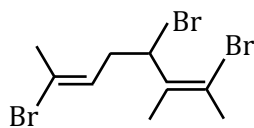
[JEE (Main) 2024]



COI155

44. Total number of stereo isomers possible for the given structure:

[JEE (Main) 2024]



- (1) 8 (2) 2 (3) 4 (4) 3

COI156

45. The isomeric deuterated bromide with molecular formula C_4H_8DBr having two chiral carbon atoms is

[JEE (Main) 2023]

- (1) 2-Bromo-1-deuterobutane
 (2) 2-Bromo-2-deuterobutane
 (3) 2-Bromo-3-deuterobutane
 (4) 2-Bromo-1-deutero-2-methylpropane

COI145

46. Total number of isomers (including stereoisomers) obtained on monochlorination of methylcyclohexane is ____.

[JEE (Main) 2022]

COI047

47. The number of stereoisomers possible for 1,2-dimethyl cyclopropane is :

[JEE (Main) 2021]

- (1) One (2) Four (3) Two (4) Three

COI045

48. The number of stereoisomers possible for a compound of the molecular formula $CH_3-CH=CH-(OH)-Me$ is :-

[AIEEE - 2009]

- (1) 4 (2) 6 (3) 3 (4) 2

COI040

EXERCISE - JEE (Advanced) PYQ

STRUCTURAL ISOMERISM

- The correct combination of names for isomeric alcohols with molecular formula $C_4H_{10}O$ is/are-
[JEE (Advanced) 2014]
(A) *tert*-butanol and 2-methylpropan-2-ol
(B) *tert*-butanol and 1, 1-dimethylethan-1-ol
(C) *n*-butanol and butan-1-ol
(D) isobutyl alcohol and 2-methylpropan-1-ol
- The number of structural isomers of C_6H_{14} is :
(A) 3 (B) 4 (C) 5 (D) 6

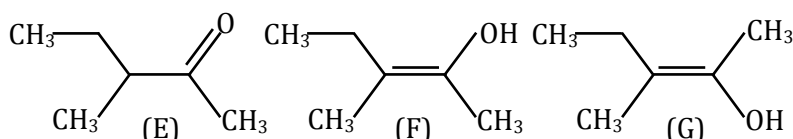
CS1061

[IIT - 2007]

CS1060

GEOMETRICAL ISOMERISM AND CONFORMATIONS

- The correct statement(s) concerning the structures E, F and G is (are)



- (A) E, F and G are resonance structures (B) E, F and E, G are tautomers
 (C) F and G are geometrical isomers (D) F and G are diastereomers

[IIT - 2008]

CG1065

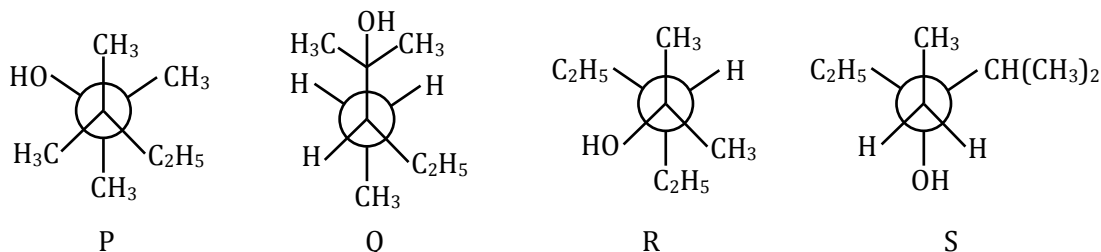
- Which of the following has the lowest dipole moment :

- (A) $\text{CH}_3\text{CH}=\text{CHCH}_3$ (trans) (B) $\text{CH}_3\text{C}\equiv\text{CCH}_3$
 (C) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$ (D) $\text{CH}_2=\text{CH}-\text{C}\equiv\text{CH}$

[IIT - 2002]

CG1063

- Newman projections **P**, **Q**, **R** and **S** are shown below :



[JEE (Advanced) 2020]

Which one of the following options represents identical molecules?

- (A) P and Q (B) Q and S (C) Q and R (D) R and S

CG1070

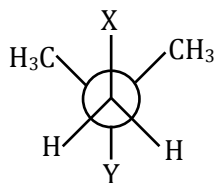
6. The bond energy (in kcal mol⁻¹) of a C–C single bond is approximately:
 (A) 1 (B) 10 (C) 100 (D) 1000

[IIT - 2010]

CGI066

7. In the Newman projection for 2,2-dimethylbutane

[IIT - 2010]



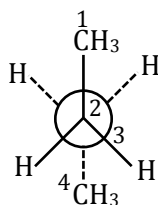
X and Y can respectively be –

- (A) H and H (B) H and C₂H₅ (C) C₂H₅ and H (D) CH₃ and CH₃

CGI067

8. If C₂ in below compound is rotated by 120° angle in anticlockwise direction along C₂–C₃, which of the following form will be produced :

[IIT - 2004]

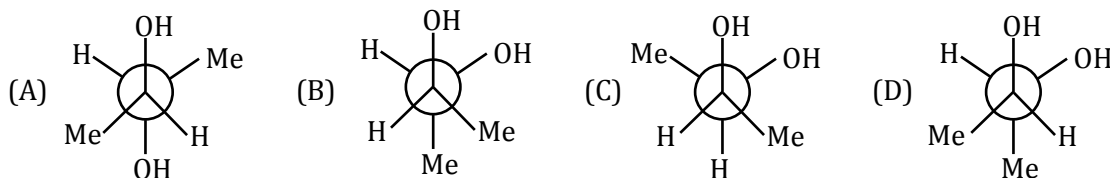


- (A) Partial eclipsed (B) Perfectly eclipsed
 (C) Perfectly staggered (D) Gauche conformation

CGI064

9. Among the following, the conformation that corresponds to the most stable conformation of *meso*-butane-2,3-diol is

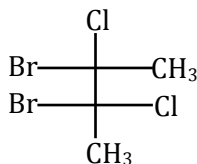
[JEE (Advanced) 2021]



CGI071

10. The total number(s) of stable conformers with non-zero dipole moment for the following compound is (are)

[JEE (Advanced) 2014]



CGI069

11. Amongst the given option, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) -

[IIT - 2011]

- (A) $\text{H}_2\text{C}=\text{C}=\text{C}=\text{CH}_2$ (B) $\text{H}-\text{C}\equiv\text{C}-\text{C}=\text{CH}_2$
 (C) $\text{H}_2\text{C}=\text{C}=\text{O}$ (D) $\text{H}_2\text{C}=\text{C}=\text{CH}_2$

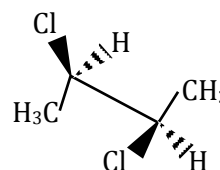
CGI068

OPTICAL ISOMERISM

12. Statement-I : Molecules that are not superimposable on their mirror images are chiral Because
Statement-II : All chiral molecules have chiral centres. [IIT 2007]
(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

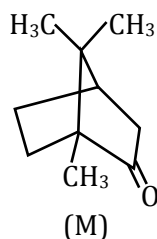
COI071

13. The correct statement(s) about the compound given below is (are) :- [IIT-2008]
(A) The compound is optically active
(B) The compound possesses centre of symmetry
(C) The compound possesses plane of symmetry
(D) The compound possesses axis of symmetry



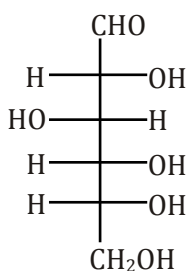
COI072

14. The total number of stereoisomers that can exist for M is : [JEE (Advanced) 2015]

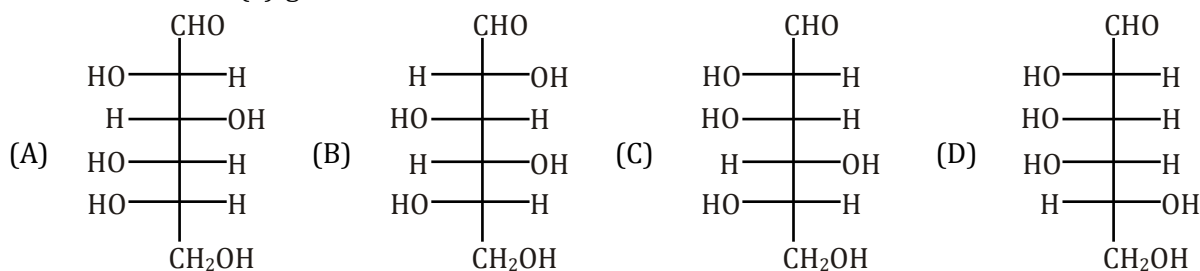


COI078

15. The structure of D-(+)-glucose is [JEE (Advanced) 2015]



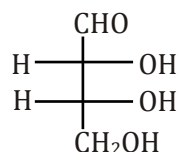
The structure of L(-)-glucose is



COI077

16. The Fischer projection of D-erythrose is shown below.

[JEE (Advanced) 2020]



D-Erythrose

D-Erythrose and its isomers are listed as **P, Q, R,** and **S** in **Column-I**. Choose the correct relationship of **P, Q, R,** and **S** with D-erythrose from **Column II**.

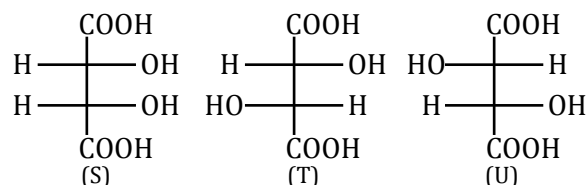
	Column-I	Column-II
(P)		1. Diastereomer 2. Identical 3. Enantiomer
(Q)		
(R)		
(S)		

(A) $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 2$ (B) $P \rightarrow 3, Q \rightarrow 1, R \rightarrow 1, S \rightarrow 2$ (C) $P \rightarrow 2, Q \rightarrow 1, R \rightarrow 1, S \rightarrow 3$ (D) $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 3, S \rightarrow 1$

CO1082

17. P & Q are isomers of dicarboxylic acid $\text{C}_4\text{H}_4\text{O}_4$. Both decolorize $\text{Br}_2/\text{H}_2\text{O}$, on heating P forms the cyclic anhydride.

Upon treatment with dilute alkaline KMnO_4 , P as well as Q could produce one or more than one from S, T and U.



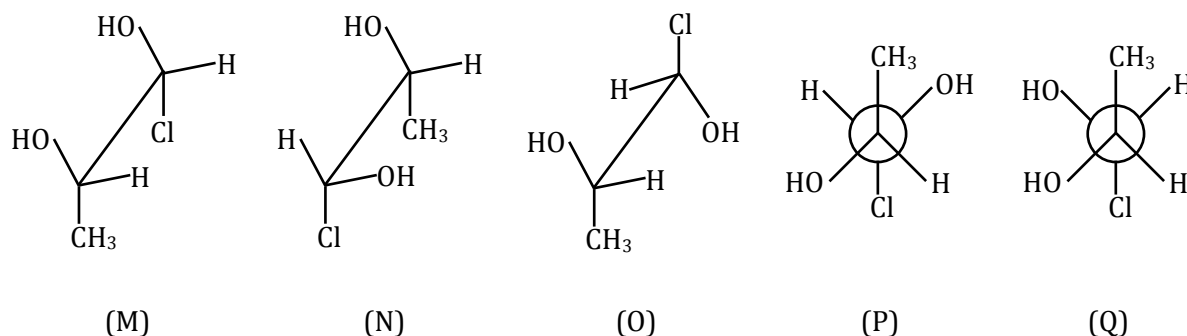
Compounds formed from P and Q are respectively.

[JEE (Advanced) 2013]

- (A) Optically active S and optically active pair (T, U)
 (B) Optically inactive S and optically inactive pair (T, U)
 (C) Optically active pair (T, U) and optically active S
 (D) Optically inactive pair (T, U) and optically inactive S

CO1076

18. Which of the given statement(s) about N,O,P and Q with respect to M is (are) correct ? [IIT 2012]



- (A) M and N are non-mirror image stereoisomers
 (B) M and O are identical
 (C) M and P are enantiomers
 (D) M and Q are identical

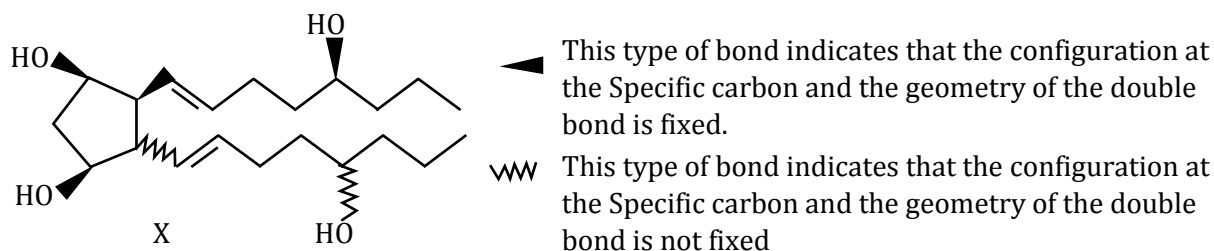
COI075

19. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is_ [JEE (Advanced) 2019]

COI081

20. For the given compound X, the total number of optically active stereoisomers is__.

[JEE (Advanced) 2018]



COI080

21. The correct statement(s) about the compound $H_3C(HO)HC-CH=CH-CH(OH)CH_3(X)$ is are :

[IIT - 2009]

- (A) The total number of stereoisomers possible for X is 6
 (B) The total number of diastereomers possible for X is 3
 (C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4
 (D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2

COI073

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	B	C	B	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	B	B	B	B	D	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	C	B	D	C	A	B	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	C	B	B	B	A	A	A	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	A	C	C	D	B	C	D	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	B	C	C	B	D	D	C	B	D

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AD	BC	ACD	ABD	BC	ABC	ABC	ABC	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AB	BCD	ACD	ABC	BD	ABC	ABC	ABC	AC	ACD
Que.	21	22	23	24	25	26	27			
Ans.	ABCD	CD	D	C	C	C	B			

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	5	3	4	5	8	4	5	7	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	6	9	6	5	3	2112	4	8	5	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	2	8	1	3	7	8	6	6
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	2	8	8	64	32	4	3.33	10.5	3.33	11.50
Que.	41	42	43	44	45	46				
Ans.	5.55	0.5	4.00	0.375	3.00	10.24				

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	4	1	4	4	4	4	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	9	8	4	2	2	3	3	1	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	6	60	3	3	1	3	4	3	5	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	6	3	3	4	2	4	3	2	42	3
Que.	41	42	43	44	45	46	47	48		
Ans.	4	2	32	1	3	12	4	1		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ACD	C	BCD	B	C	C	BD	D	B	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	BC	C	AD	2	A	C	B	ABC	10	7
Que.	21									
Ans.	AD									